

AS/NZS 1158.3.1:2020
(Incorporating Amendment No. 1)



Australian/New Zealand Standard™

Lighting for roads and public spaces

Part 3.1: Pedestrian area (Category P) lighting — Performance and design requirements



This Joint Australian/New Zealand Amendment was prepared by Joint Technical Committee LG-002, Lighting for Roads and Public Spaces. It was approved on behalf of the Council of Standards Australia on 11 February 2020 and by the New Zealand Standards Approval Board on 05 February 2020.

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- Astronomical Society of Australia
- Australian Industry Group
- Australian Local Government Association
- Centre for Pavement Engineering Education
- CIE Australia
- Consumers Federation of Australia
- Department of Planning, Transport and Infrastructure, SA
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- Lighting Council Australia
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- Main Roads Western Australia
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This Standard was issued in draft form for comment as DR AS/NZS 1158.3.1:2018.

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Lighting for roads and public spaces

Part 3.1: Pedestrian area (Category P) lighting — Performance and design requirements

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Preface

This Standard was prepared by the Joint Standards Australia/Standards New Zealand Committee LG-002, Lighting for Roads and Public Spaces, to supersede AS/NZS 1158.3.1:2005, *Lighting for roads and public spaces*, Part 3.1: *Pedestrian area (Category P) lighting—Performance and design requirements*.

[A₁] This Standard incorporates Amendment No. 1 (October 2024). The start and end of changes introduced by the Amendment are indicated in the text by tags including the Amendment number 1. **[A₁]**

This Standard forms Part 3.1 of the AS/NZS 1158 series, which covers lighting schemes for the generality of roads and outdoor public areas.

The AS/NZS 1158 series consists of the following:

AS/NZS 1158.0, Lighting for roads and public spaces, Part 0: Introduction

AS/NZS 1158.1.1, Lighting for roads and public spaces, Part 1.1: Vehicular traffic (Category V) lighting—Performance and design requirements

AS/NZS 1158.1.2, Lighting for roads and public spaces, Part 1.2: Vehicular traffic (Category V) lighting—Guide to design, installation, operation and maintenance

AS/NZS 1158.2, Lighting for roads and public spaces, Part 2: Computer procedures for the calculation of light technical parameters for Category V and Category P lighting

AS/NZS 1158.3.1, Lighting for roads and public spaces, Part 3.1: Pedestrian area (Category P) lighting—Performance and design requirements (this Standard)

AS/NZS 1158.4, Lighting for roads and public spaces, Part 4: Lighting of pedestrian crossings

AS/NZS 1158.5, Lighting for roads and public spaces, Part 5: Tunnels and underpasses

AS/NZS 60598.2.3, Luminaires, Part 2.3: Particular requirements—Luminaires for road and street lighting (IEC 60598-2-3, Ed.3.1 (2011) MOD)

SA/SNZ TS 1158.6, Lighting for roads and public spaces, Part 6: Luminaires—Performance

The significant technical changes that have been made in this Standard in relation to the 2005 edition include the following:

- (a) Separation of previous Table 2.6 into the differing requirements for lighting of ‘local roads’ and ‘cyclist paths’.
- (b) Clarification of glare requirements for high intensity discharge luminaires and new requirements for SSL light sources.
- (c) Additional and revised layout rules for typical road layouts.
- (d) Additional information relating to SSL light sources.
- (e) Additional requirements for surround illuminance.
- (f) Updated data on the specification of minimum ingress protection requirements for various maintenance factors and requirements regarding assumed and actual maintenance regimes and maintenance intervals.
- (g) Additional lighting level for lighting of external car parking areas.
- (h) New requirements relating to energy measures if required.
- (i) Additional option for specifying minimum environmental spill levels for local roads.

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This joint Standard is intended to be applied in its entirety in Australia and New Zealand; however, a number of differences exist with respect to the requirements that apply in each country. These are indicated by the qualification 'In Australia' or 'In New Zealand', or similar.

The terms 'normative' and 'informative' have been used in this Standard to define the application of the appendix to which they apply. A 'normative' appendix is an integral part of a Standard, whereas an 'informative' appendix is only for information and guidance.

Statements expressed in mandatory terms in footnotes to tables are deemed to be requirements of this Standard.

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Foreword

The performance criteria for road and public space lighting schemes can include any or all of the three basic aims of—

- (a) facilitation of safe movement;
- (b) reduction of the fear of crime at night; and
- (c) contributing to the amenity of an area through increased aesthetic appeal.

However, as the most common primary aim of a scheme is that of safe movement of people, the series divides road and public space lighting into two broad categories — Category V lighting and Category P lighting, as defined in AS/NZS 1158.0.

This Standard sets out performance and design requirements for Category P lighting schemes having regard to the safe movement of pedestrians, degree of activity (of pedestrians and vehicles), the fear of crime and the need to enhance the amenity of the locality. It should be read in conjunction with AS/NZS 1158.2, which sets out calculation procedures.

Category P lighting is acknowledged to be an effective counter measure to the fear of crime.

The effects that blue light can have on human melatonin suppression and circadian rhythm disruption have begun to be explored extensively in the literature. However, there seems to be no definitive levels of amount, duration and timing which would give guidance to this document.

For each lighting subcategory described in this Standard, the light technical parameters (LTPs) and their prescribed values are both necessary and sufficient for the particular application. Conformance to this Standard will be achieved by meeting all the required values of the LTPs for the designated subcategory. A higher quality of lighting scheme can be achieved within a subcategory by, for example, increasing the minimum level of uniformity or reducing the glare allowable, or both.

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Lighting for roads and public spaces

Part 3.1: Pedestrian area (Category P) lighting — Performance and design requirements

Section 1 Scope and general

1.1 Scope and application

1.1.1 Scope

This Standard specifies performance and design requirements for Category P lighting schemes as described in AS/NZS 1158.0. It also specifies the luminaire data and other data that is needed to facilitate the lighting design and the assessment of conformance to the requirements of this Standard.

This Standard assigns various levels of required performance based on an assessment of degree of activity, fear of crime and the required aesthetic appeal (i.e. 'amenity'). In achieving the required performance, attention is paid to minimizing adverse effects on the night time environment.

This Standard does not apply to roads in which one or a number of luminaires may be installed at irregular intervals on an otherwise unlit road, nor to additional luminaires installed in lit roads for reasons of increased security, including instances where such luminaires are typical of those used in Category P lighting schemes.

This Standard does not cover lighting intended specifically as security lighting for building exteriors, nor does it cover lighting for video surveillance. However, where video surveillance is provided, the lighting principles and performance requirements expressed in this Standard still apply.

This Standard does not cover lighting requirements for 'wayfinding' or 'waypoint' lighting where lights are intended to provide a guide rather than to provide a particular level of illumination.

This Standard does not apply in Australia to public transport areas where the mandatory requirements of the *Disability Discrimination Act 1992 (Cth)* and subordinate instruments including the *Disability Standards for Accessible Public Transport 2002 (Cth)* (DSAPT) and the *Disability (Access to Premises – Buildings) Standards 2010 (Cth)* (Premises Standards) are applicable. See [Clause 2.1](#).

This Standard is intended to be read in conjunction with AS/NZS 1158.0, AS/NZS 1158.2, AS/NZS 1158.1.1, AS/NZS 60598.2.3 and SA/SNZ TS 1158.6, which specify related provisions that may require consideration.

1.1.2 Application

Subject to the requirements of applicable laws, the choice of whether to install a scheme of road or public space lighting in conformance to this Standard and, if so, which subcategory of lighting is appropriate, rests with the client (usually the applicable road controlling authority). This decision is typically based on factors such as night-time pedestrian traffic flows and other patterns of use.

1.2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document.

NOTE Documents for informative purposes are listed in the Bibliography. Following the informative referenced documents, related documents that are considered useful sources of information on the subject of this Standard are provided.

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AS 1742.13, *Manual of uniform traffic control devices, Part 13: Local area traffic management*

AS ISO/IEC 17025, *General requirements for the competence of testing and calibration laboratories*

AS/NZS 1158.0, *Lighting for roads and public spaces, Part 0: Introduction*

AS/NZS 1158.1.1, *Lighting for roads and public spaces, Part 1.1: Vehicular traffic (Category V) lighting—Performance and design requirements*

AS/NZS 1158.1.2, *Lighting for roads and public spaces, Part 1.2: Vehicular traffic (Category V) lighting—Guide to design, installation, operation and maintenance*

AS/NZS 1158.2, *Lighting for roads and public spaces, Part 2: Computer procedures for the calculation of light technical parameters for Category V and Category P lighting*

AS/NZS 4282, *Control of the obtrusive effects of outdoor lighting*

AS/NZS 60598.2.3, *Luminaires, Part 2.3: Particular requirements — Luminaires for road and street lighting (IEC 60598-2-3, Ed. 3.1 (2011) MOD)*

CIE S 025/E,¹⁾ *Test Method for LED Lamps, LED Luminaires and LED Modules*

IESNA LM-79-19, *Electrical and Photometric Measurements of Solid-State Lighting Products*

IESNA LM-80-15, *IES Approved Method: Measuring Luminous Flux and Color Maintenance of LED Packages, Arrays and Modules*

IESNA LM-84-14, *Approved Method for Measuring Luminous Flux and Color for Maintenance of LED Lamps, Light Engines and Luminaires*

IESNA TM-21-11, *Projecting Long Term Lumen Maintenance of LED Light Sources*

NEW ZEALAND TRANSPORT AGENCY. *Traffic Control Devices Manual*

1.3 Definitions

For the purpose of this Standard, the definitions given in AS/NZS 1158.0 and the following apply.

1.3.1

competent person

a person who has acquired, through education, training, qualification or experience or a combination of these, the knowledge and skill enabling that person to perform the task required as determined by the client

1.3.2

initial lamp lumens (for non-SSL luminaires)

the total luminous flux emitted by a light source after 100 h of operation

1.3.3

initial luminaire lumens (for SSL luminaires)

the total luminous flux emitted by the SSL luminaire as photometered according to CIE S 025/E or IESNA LM-79-19

1.3.4

solid state lighting

SSL

a group of devices that embody p-n junctions that emit visible radiation when excited by an electric current. They can be further defined by the type of technology used, i.e. light-emitting diodes (LEDs), organic light-emitting diodes (OLEDs) or polymer light-emitting diodes (PLEDs)

1) International Commission on Illumination, Vienna.

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1.3.5 **car parks outdoor** **1.3.5.1**

off-road car park

area either on grade or on a structure that is on a separate land title to the road reserve and is designed for the parking of vehicles and includes defined entries and exits and access road(s)

1.3.5.2 **on-road car parking**

parking spaces on the road reserve including parallel, angle and median parking and parking for the disabled ^(A1)

1.3.6 **S/P ratio**

R_{sp}

ratio of the luminous output of a source evaluated according to the CIE scotopic spectral luminous efficiency, $V'(\lambda)$, to the luminous output evaluated according to the CIE photopic spectral luminous efficiency, $V(\lambda)$

Note 1 to entry: The S/P ratio is expressed by

$$R_{sp} = \frac{K'_m \int_0^{\infty} s(\lambda) V'(\lambda) d\lambda}{K_m \int_0^{\infty} s(\lambda) V(\lambda) d\lambda}$$

where

K'_m = approximately 1 700 lm·W⁻¹ is the maximum luminous efficacy for scotopic vision

K_m = approximately 683 lm·W⁻¹ is the maximum luminous efficacy for photopic vision

$s(\lambda)$ = the relative spectral distribution of the source, and λ is the wavelength

Note 2 to entry: The S/P ratio has unit one.

[SOURCE: CIE S 017/E:2020, Entry 17-21-113] ^(A1)

1.4 Symbols and abbreviations

For the purpose of this Standard, the symbols and abbreviations given in AS/NZS 1158.0 apply.

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Section 2 Objectives and determination of lighting subcategories

2.1 Road and public space classification

[Figure 2.1](#) shows representative land use types and associated road and public space classifications together with the applicable lighting categories and subcategories.

NOTE Selection of the appropriate lighting subcategory for a road or public space is governed by [Tables 2.1](#) to [2.5](#) for Category P areas and AS/NZS 1158.1.1 for Category V areas.

The exact names of the various road types and their associated operating characteristics may vary between relevant authorities. In addition, the layout of roads in older suburbs may not conform to the schematic layout shown in [Figure 2.1](#). In particular, the local roads and some arterials might coalesce into a grid arrangement of roads. It is therefore important that the documentation for the lighting design clearly states the road or public space type and associated lighting subcategory that have been assigned.

When determining the applicable lighting category and subcategory each separate element needs to be assessed by considering its own particular operational characteristic. Special attention should be given to the likely night time usage. This may differ with respect to traffic density or the presence of pedestrians to that existing during the day. For example, a neighbourhood collector road may have relatively low traffic volumes with pedestrian activity during the day, but at night be used by heavy vehicles servicing industrial or retail precincts.

Public spaces within public transport boundaries (including, but not limited to, walkways, pedestrian crossings, disability car parking spaces) are subject to Commonwealth Legislation. Designers should refer to the DSAPT and the Premises Standards, which contain specific illumination requirements, and to AS 1428.2:1992. Further information on the implementation on both of the DSAPT and the Premises Standards is available from their guidelines. Otherwise public spaces surrounding public transport boundaries shall be classified in accordance with this Standard.

2.2 General objectives

The major purposes of the lighting covered in this Standard are to assist pedestrians to orientate themselves and detect potential hazards, and to reduce fear of crime while protecting the integrity of the night time environment through control of obtrusive light and glare. The lighting may also be used to enhance the amenity of the location and should be designed to minimize any obtrusive effects.

The lighting, with certain exceptions, is not designed to provide drivers with enhanced and adequate visibility if motor vehicle traffic is present at the location. Drivers need to use the vehicle headlights for visibility. The exceptions are LATM devices and where there is interactive pedestrian and vehicular activity present in designated areas, e.g. transport interchanges, car parks.

The provision of road lighting should not compromise other road safety objectives, e.g. by creating a potential collision hazard through the use of road lighting poles/columns that are of an inappropriate type or that are inappropriately placed.

NOTE Refer to appropriate appendix of AS/NZS 1158.1.2 for guidance on the selection and locations of road lighting columns and poles.

2.3 Design objectives

2.3.1 Principal objectives

The principal design objectives for Category P lighting are to provide the following:

- (a) Illuminance and uniformity of illuminance over the designated area to a specified level.
- (b) Glare control to a specified level.

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- (c) Limitation of upward light from luminaires to a specified level.
- (d) Limitation to a specified level of the obtrusive light into properties that abut the lit area.
- (e) A lighting scheme that conforms to this Standard at all times during each maintenance cycle over the life of an installation.
- (f) Minimization of energy consumption.

The specified levels of illuminance depend on the particular lighting subcategory involved (see [Tables 2.1](#) to [2.5](#) inclusive).

These objectives are realized by a combination of the choice of relevant lighting subcategory and appropriate design procedure, leading to luminaire light output distribution and lighting scheme geometry appropriate to the configuration and dimensions of the location being lit.

Road surface reflection properties are not taken into account in the design of Category P lighting for roads.

2.3.2 Other considerations

Other aspects of a road or public space lighting scheme will affect the initial and long-term performance quality. These include the following:

- (a) Reliability and maintainability over the service life of the selected hardware components.
- (b) Energy efficiency.
- (c) Ability to complement or enhance the visual amenity of the illuminated route or area by the aesthetics of the installation, including the use of light with good colour rendering properties (see Clause 3.3).
- (d) Life cycle costs of the scheme.

2.4 Lighting subcategories and their application

For the purpose of this Standard the lighting subcategories set out in the following tables are:

- (a) For roads in local areas (PR)—[Table 2.1](#).
- (b) For pathways (PP)—[Table 2.2](#).
- (c) For public activity areas (excluding car parks) (PA)—[Table 2.3](#).
- (d) For connecting elements (PE)—[Table 2.4](#).
- (e)  For outdoor off-road car parks (including roof-top car parks) (PC)—[Table 2.5](#) .

[Tables 2.1](#) to [2.5](#) set out the general application and operating characteristics of the roads or public spaces to which the lighting subcategories apply. [Figure 2.1](#) illustrates indicative lighting subcategories for a range of road and public space types on a schematic locality plan.

The concept of using relative terms is employed since it would be both unrealistic and impractical to apply one numerical set of values, for each of the selection criteria, to the lighting level hierarchy to accommodate the relevant requirements of each town, city, etc., across both Australia and New Zealand.

The terms 'mixed', 'low', 'medium', 'high', 'very high' and 'moderate' are provided to enable individual road authorities to develop their own specific lighting hierarchy.

The appropriate subcategory for a particular road or public space should be a matter for determination in consultation with the client (usually the applicable road controlling authority). When a geographically specific road lighting scheme incorporates road or roads controlled by more than one road authority,

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then it will be necessary to consult those authorities in order to confirm the subcategory of road lighting which is to be specified for the road lighting scheme.

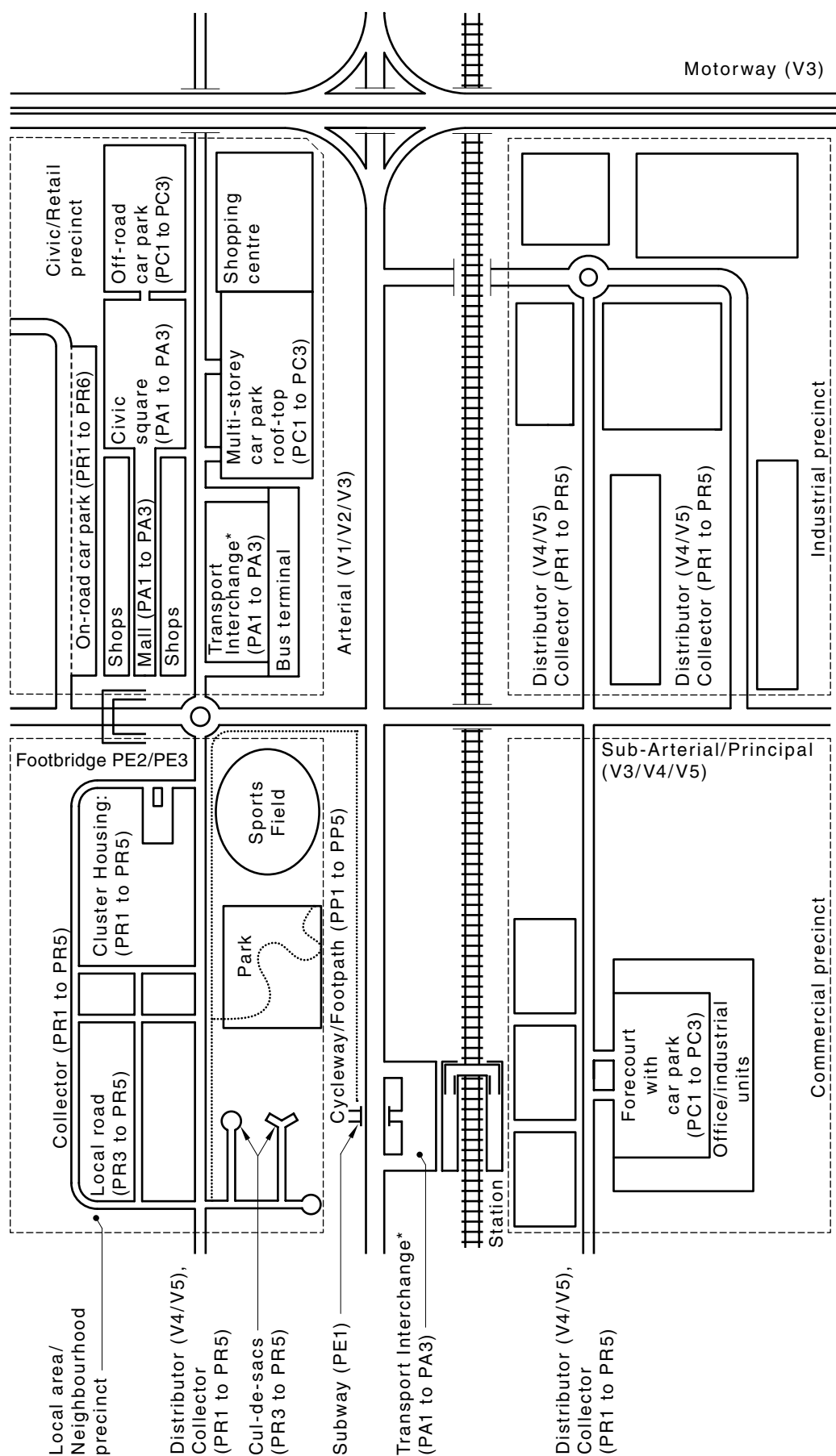
As the operating characteristics of a road or public space are not always constant throughout the operating period due to changes in vehicle volume or pedestrian/cycle activity it is appropriate for a lighting scheme to have more than one applicable lighting subcategory during normal operation.

NOTE 1 [Appendix A](#) provides information on the criteria used in [Tables 2.1](#) to [2.3](#) to determine the applicable lighting subcategory, based on the fear of crime or the need to enhance the amenity of the locality. The appendix also includes an example of the selection procedure.

NOTE 2 See [Appendix B](#) for a sample design brief.

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* See Clause 1.1 regarding Disability Discrimination Act 1992 (Cth) otherwise (PA1/PA2/PA3)

Figure 2.1 — Example of road and public space types and indicative lighting categories and subcategories

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Table 2.1 — Lighting subcategories for road reserves in local areas

1	2	3	4	5	6
Type of road or pathway		Selection criteria ^{a,b}			Applicable lighting subcategory ^{c,d}
General description	Basic operating characteristics	Pedestrian/cycle activity	Fear of crime	Need to enhance amenity	
Collector roads or non-arterial roads which collect and distribute traffic in an area, as well as serving abutting properties	Mixed vehicle and pedestrian traffic	N/A	High	N/A	PR1
		High	Medium	High	PR2
		Medium	Low	Medium	PR3 ^f or PR4 ^f
		Low	Low	Low	PR5
Local roads or streets used primarily for access to abutting properties, including residential, commercial and industrial precincts		N/A	High	N/A	PR1
		High	Medium	High	PR2
		Medium	Low	Medium	PR3 ^f or PR4 ^f
		Low	Low	Low	PR5
Common area, forecourts of cluster housing		N/A	N/A	N/A	PR6 ^e
		N/A	High	N/A	PR1
		High	Medium	High	PR2
		Medium	Low	Medium	PR3 ^f or PR4 ^f
		Low	Low	Low	PR5

^a The selection criteria of Columns 3 to 5 should be separately evaluated. The highest level of any of the selection criteria that is deemed appropriate for the road will determine the applicable lighting subcategory.

^b See [Appendix A](#) for guidance on choosing the applicable level of each selection criteria for the environment and purpose of a lighting scheme.

^c All lighting subcategories apply across the whole of the road reserve width, including the footpath.

^d Where there is a significant fear of crime or where required by the relevant authority, then, for enhanced lighting of the formed pathways, see [Table 2.2](#).

^e Use of subcategory PR6 shall be discretionary. Generally, subcategory PR6 is only applied to the replacement of existing luminaires installed on existing electricity distribution poles or for the initial application of a lighting scheme where the cost to reconfigure these poles limits or precludes conformance to subcategory PR4 and PR5 respectively. However, it is recognized that, for some authorities, there may be some specific lighting tasks where subcategory PR5 could be deemed to be excessive in terms of providing adequate level of service and meeting with community expectations. In this case subcategory PR6 may be used.

[A1] [Text deleted.] [A1]

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Table 2.2 — Lighting subcategories for pathways

1	2	3	4	5
Type of pathway		Selection criteria ^{a,b,c}		Applicable lighting subcategory
General description	Basic operating characteristics	Pedestrian/cycle activity	Fear of crime	
Pedestrian, cycle, etc. orientated pathway, e.g. footpaths, including those along local roads ^d and arterial roads ^e , walkways, lanes, park paths, cyclist paths	Pedestrian, cycle, mobility devices, scooter, etc. traffic only	N/A	High	PP1 ^c
		High	Medium	PP2 ^c
		Medium	Medium	PP3
		Medium	Low	PP4
		Low	Low	PP5
^a The selection criteria of Columns 3 to 4 should be separately evaluated. The highest level of any of the selection criteria that is deemed appropriate for the pathway will determine the applicable lighting subcategory. ^b See Appendix A for guidance on choosing the applicable level of each selection criteria for the environment and purpose of a lighting scheme. ^c Where there are vertical surfaces of high reflectance (e.g. light coloured walls bordering on an alleyway) alongside the pathway, the next lower lighting subcategory may be selected. ^d Where the footpath is along a local road and subcategory PP1 or PP2 is selected, the light technical parameters for that subcategory should only apply to the formed footpath. ^e Footpaths associated with arterial roads are deemed not to require separate lighting provided that— (a) the road is lit to at least the applicable level of Category V lighting conforming to AS/NZS 1158.1.1; and (b) the footpath is unshaded, e.g. there are no substantially continuous building awnings, trees (refer to AS/NZS 1158.1.2) and the footpath is contiguous with the roadway. [A1]				

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Table 2.3 — Lighting subcategories for public activity areas (excluding car parks)

1	2	3	4	5	6
Type of area or activity		Selection criteria ^{a,b}			Applicable lighting subcategory
General description	Basic operating characteristics	Night time vehicle movements	Fear of crime	Need to enhance amenity	
Areas primarily for pedestrian use, e.g. city, town, suburban centres, including outdoor shopping precincts, malls, open arcades, town squares, civic centres	Generally pedestrian movement only	N/A	High	High	PA1
		Medium	Medium	Medium	PA2
		Low	Low	N/A	PA3
Transport terminals and interchanges, service areas ^c	Mixed pedestrian and vehicle movement	High	High	High	PA1
		Medium	Medium	Medium	PA2
		Low	Low	N/A	PA3

^a The selection criteria of Columns 3 to 5 should be separately evaluated. The highest level of any of the selection criteria that is deemed appropriate for the area type will determine the applicable lighting subcategory.

^b See [Appendix A](#) for guidance on choosing the applicable level of each selection criteria for the environment and purpose of the lighting scheme.

^c See [Clause 1.1.1](#).

NOTE See [Table 2.5](#) for lighting subcategories applicable to outdoor car parks, including roof-top car parks.

Table 2.4 — Lighting subcategories for connecting elements

Type of area	Applicable lighting subcategory
Subways, including associated ramps or stairways	PE1
Steps and stairways, ramps, footbridges, pedestrian ways	PE2
Ramps and footbridges associated with low use pathways (e.g. in parks and reserves)	PE3
NOTE Subways are listed as a separate subcategory because of a fear of crime.	

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Table 2.5 —  Lighting subcategories for outdoor off-road car parks (including roof-top car parks) 

1	2	3	4
Type of area	Selection criteria ^{a,c}		
	Night time vehicle and/or pedestrian movements	Fear of crime	Applicable lighting subcategory ^b
Parking spaces, aisles and circulation roadways	High	High	PC1
	Medium	Medium	PC2
	Low	Low	PC3
Designated parking spaces specifically intended for people with disabilities	N/A	N/A	PCD
 Designated areas for pedestrians to cross circulation roadways and access roadways 	N/A	N/A	PCX

^a The selection criteria of Columns 2 to 4 should be separately evaluated. The highest level of any of the selection criteria that is deemed appropriate for the area type will determine the applicable lighting subcategory.

^b Providing a lighting scheme that meets the requirements of more than one subcategory by the use of switching is permitted.

^c Consider the use of adaptive lighting controls for variable night time utilization.

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Section 3 Light technical parameters for each lighting subcategory

3.1 Light technical parameters

3.1.1 General

The design objectives outlined in [Clause 2.3](#) are formally specified in terms of various light technical parameters. These parameters are divided as follows:

- (a) Parameters that relate to the attainment of the required level of lighting performance.
- (b) Parameters that limit the adverse effects of the lighting on—
 - (i) users of the lit space, e.g. pedestrians, vehicle drivers (i.e. by the control of glare);
 - (ii) night sky viewing conditions (i.e. by reducing consequent sky glow); and
 - (iii) occupants of properties that abut the lit area (i.e. by the control of obtrusive light).

Details of the applicable parameters are set out in [Clauses 3.1.2](#) and [3.1.3](#) [the mathematical definitions of these light technical parameters (LTPs) are set out in AS/NZS 1158.2].

The values for these parameters shall be as specified in Tables 3.3 to 3.7 and are the minimum permissible, except for the parameter U_{E2} for which the values given are the maximum permissible. These requirements apply throughout each maintenance cycle over the life of the installation.

When adaptive lighting is used, each dimming level shall be such that it complies with all the requirements of a relevant lighting category. The control system shall be failsafe in that on a failure of any element of the control system, the luminaires shall revert back to their initial undimmed level.

The design procedures necessary to determine conformance to the required values of the light technical parameters shall be as specified in [Section 4](#).

3.1.2 Lighting performance parameters

The performance of Category P lighting shall be assessed in terms of the light technical parameters in Table 3.1:

Table 3.1 — Light technical parameters (LTP)

Parameter	Symbol
Average horizontal illuminance	$\bar{E}_h$
Point horizontal illuminance	E_{Ph}
Point horizontal illuminance within the pathway surround	E_{PhS}
Illuminance uniformity Category P	U_{E2}
Point vertical illuminance	E_{Pv}
Discomfort glare index	DGI_p A1

The applicable values for these four parameters shall be as set out in Tables 3.3 to 3.7 for the lighting subcategories established in accordance with [Clause 2.4](#).

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3.1.3 Adverse effects of lighting

3.1.3.1 General

The adverse effects of the lighting system shall be controlled by application of the light technical parameters set out in [Clauses 3.1.3.2 to 3.1.3.4](#).

3.1.3.2 Glare

Glare from the luminaires shall be mitigated by the specification of maximum values of the luminous intensity or of the average controlled luminance in specified directions, appropriate to the luminaire type (see [Clause 3.7](#)).

3.1.3.3 Upward waste light

The amount of light emitted by the luminaires in directions above the horizontal shall be controlled by the specification of maximum values for the upward waste light ratio (*UWLR*).

The type of luminaires used and the way that they are installed shall ensure conformance to the requirements of Table 3.10. The limiting value of *UWLR* in Table 3.10 shall not be exceeded for the selected luminaire when mounted with the tilt angle, where applicable, specified in the design.

NOTE 1 [Section 5](#) requires that luminaire manufacturers state the maximum tilt angle at which the luminaire satisfies the requirement in Table 3.10.

NOTE 2 The *UWLR* typically increases with tilt.

3.1.3.4 Spill light into abutting properties

For local roads provided with lighting subcategories PR3 to PR6 spill light can be advantageous in lighting the surrounds. If deemed beneficial, a minimum level of 50 % of E_{ph} at a distance of 3 m inside abutting properties should prevent any undesirable sharp reduction in illumination immediately inside the property boundaries.

However, if it is deemed necessary by the relevant authority that a limit is required on the amount of spill light, refer to AS/NZS 4282.

[A1] NOTE Off-road parking installations are not considered as part of public lighting by AS/NZS 4282 and are within the formal scope of AS/NZS 4282. **[A1]**

3.1.3.5 Impact on wildlife

Spill light from roads and pathways may have negative impacts upon wildlife species sensitive to light. In some cases where environmental habitat or migratory pathways of these species is adjacent or nearby, relevant regulation and regulatory authorities may require that spill light be limited in order to avoid or minimize negative impacts.

In Australia, this, in particular, relates to instances where threatened or migratory species or threatened ecological communities listed under the Australian Government, Department of Environment and Energy — *Environment Protection and Biodiversity Conservation Act 1999 (Cth)*, for which light is listed as a threat, occur in the vicinity of the lighting.

New Zealand users will find relevant details at the following link:

<https://www.doc.govt.nz/nature/biodiversity/biodiversity-new-zealandresources/biodiversity-references/laws-and-regulations>

Further guidance on addressing the impact of spill light upon wildlife is available in AS/NZS 4282.

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3.2 Light source requirements

3.2.1 General

The choice of light source should be based on an analysis of all the factors relevant to the particular application, including aesthetics, environmental factors, light source mortality, lumen depreciation, cost and energy use.

For correlated colour temperature (CCT) and colour rendering index (CRI) for light sources, refer to SA/SNZ TS 1158.6.

NOTE The performance and characteristics of a light source are not fully defined by CCT and CRI metrics and consideration should be taken of the spectral power distribution (SPD) of a light source.

3.2.2 Design lumen value

Except as detailed below the initial (lamp or luminaire) lumen value, as conventionally evaluated using the CIE 1924 Standard Observer for photopic vision (i.e. the $V(\lambda)$ function), shall be used in the calculations.

The exceptions are—

- (a) for lighting road subcategories PR3, PR4, PR5, PR6; and
- (b) for pathways and cyclist subcategories of PP4 and PP5.

For the subcategories listed above the lumen multipliers shall be used to derate the effective lumens when using light sources with an S/P ratio < 1 and/or a CCT $< 2\,500$ K.

Lumen multipliers shall be derived according to either [3.2.2 \(a\)](#), [3.2.2 \(b\)](#) or [3.2.2 \(c\)](#). The lumen multipliers determined shall be used to calculate the Design Lumen value using the following formula:

Design lumen value = Initial lumen value $\times$ lumen multiplier

- (i) Using S/P ratio—
 - (A) for sources with S/P ratio ≥ 1 , no derating (lumen multiplier = 1); and
 - (B) for sources with S/P ratio < 1 , derate using the formula—

$$\text{Lumen multiplier} = 0.65 \times \text{S/P ratio} + 0.35$$
- (ii) Using CCT, if the S/P ratio is not available—
 - (A) for sources with CCT $\geq 2\,500$ K, no derating (lumen multiplier = 1);
 - (B) for sources with CCT between 1 900 K and 2 500 K, derate using a lumen multiplier of 0.85; and
 - (C) for sources with CCT $\leq 1\,900$ K, derate using a lumen multiplier of 0.50.
- (iii) For sodium vapour lamps—
 - (A) for high-pressure sodium lamps, derate using a lumen multiplier of 0.75; and
 - (B) for low-pressure sodium lamps, derate using a lumen multiplier of 0.50.

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Lumen multipliers of a value greater than 1.0, such as for sources with high S/P ratio, shall not be used.

NOTE 1 These requirements are included to compensate for the shift in sensitivity of the eye towards the blue end of the spectrum at low light levels, resulting in decreased sensitivity of the eye to yellow/red light and increased sensitivity to blue light, and also for the generally poorer colour rendering of sources with lower S/P ratio. The low brightness of the scene will be further exacerbated by the low reflectance of any surrounding foliage to yellow light.

NOTE 2 The values for high-pressure sodium and low-pressure sodium lamps are given for historical purposes.

Refer to the following references:

- (A) Fisher, A.J., "Can a lumen be less than a lumen so that white light is better than yellow in road lighting? Part two." *Lighting*. 2001. **21** pp. 42-46.
- (B) Bergen, T., "Recommendation on derating LED sources in the lower P category roadway lighting levels", Centre for Pavement Engineering Education, 4/10/2022, <https://www.pavementeducation.edu.au/info/documents/publications/publications/recommendation-on-derating-led-sources-in-the-lower-p-category-roadway-lighting-levels>.
- (C) CIE 191:2010, Recommended system for Mesopic photometry based on visual performance. International Commission on Illumination, Vienna, 2010. ^{A1}

3.3 ^{A1} [Text deleted.] ^{A1}

3.4 Energy performance assessment and reporting

Where energy performance assessment and reporting is required by road authorities the methods shall be as set out in [Appendix F](#).

The methods outlined in [Appendix F](#) align with the principles in EN 13201-5 which define international practice.

The energy performance of a lighting scheme is determined by the interaction and combined effects of the—

- (a) luminous efficacy and photometric distribution of the selected luminaire;
- (b) appropriateness of the lighting design for the scheme layout and geometry; and
- (c) operation of the lighting system in practice.

The ability to objectively quantify and compare the total systemic energy impacts of road lighting scheme scenarios at the planning stages provides the facility for the lighting designer to undertake tuning and optimization of various options to evaluate the energy use of a proposed lighting scheme while meeting the required light technical parameters.

The methods in [Appendix F](#) accommodate lighting levels as dynamic and changing parameters rather than only as fixed values. A variable light level and timing facility allows for the application of adaptive lighting techniques such as constant light output (CLO), scheduled control and real-time control enabled by smart control devices and internet based central management systems (CMS).

The methods in [Appendix F](#) provide rules for the structured description, calculation and presentation of annualized energy performance indicators for road, pathway and public space lighting schemes.

NOTE This will facilitate optimization at the design stage and for ongoing periodic auditing purposes. This will allow for structured and harmonized comparisons of the energy performance implications of—

- (a) various equipment technologies;
- (b) design approaches such as luminaire optics and column layouts;

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- (c) adapting at times of low pedestrian use, e.g. after midnight, using sensors to adapt between lighting categories on approach of pedestrians; and
- (d) operating of dimming/brightening adaptive control schedules or sensing.

Application of the methods in [Appendix F](#) will provide a standardized framework to design, calculate, benchmark or audit the energy performance of both existing and proposed lighting schemes in a systemic and objective manner for the purposes of benchmarking, continuous improvement or as part of progressive infrastructure procurement processes.

[Appendix F](#) introduces the measures of—

- (i) the power density indicator (PDI); and
- (ii) the annual energy consumption indicator (AECI).

3.5 Maintenance of light technical parameters

3.5.1 General

Significant reductions from the initial values of the light technical parameters will occur in service, primarily as a result of a gradual depreciation in light source lumen output, an accumulation of dirt and ageing of the optical transmitting materials and/or reflecting surfaces of the luminaire. Recognizing that this depreciation occurs, [Tables 3.3 to 3.7](#) require conformance to maintained values of $\bar{E}_h$, E_{ph} and E_{pv} by the application of a maintenance factor, see [Clauses 3.5.2 to 3.5.5](#).

An appropriate maintenance regime shall be developed and documented for the design which will ensure that the values of $\bar{E}_h$, E_{ph} and E_{pv} will not fall below the applicable values for these parameters throughout any maintenance cycle in the life of the installation.

3.5.2 Luminaire dirt depreciation factor (LDD)

The value of the luminaire optical degradation factor associated with the pollution category and cleaning interval to be assumed for the calculation of maintained values $\bar{E}_h$, E_{ph} and E_{pv} , shall not exceed the applicable value in this Clause. The IP rating of the luminaire shall not be less than IP65. If the control gear is IP65, the control gear/wiring connection chamber may be IP24.

[Table 3.2](#) gives values of luminaire maintenance factors for luminaires with IP ratings of IP65 and above by environmental zone and cleaning interval. These values shall be used in Category P lighting system design. The effects of light source lumen depreciation are not included in the luminaire dirt depreciation (LDD) factor.

NOTE Some additional information is given in AS/NZS 1158.1.2, however [Table 3.2](#) supersedes the LDD factors for IP6X luminaires in AS/NZS 1158.1.2:2010 Table 14.4. The LDD factors for IP54 luminaires are given in AS/NZS 1158.1.2:2010 Table 14.4.

Table 3.2 — LDD factors

Environmental zone	LDD factor			
	Cleaning frequency			
	36 months	48 months	60 months	72 months
Rural	0.95	0.94	0.93	0.92
Urban	0.90	0.88	0.86	0.84

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3.5.3 Light source lumen depreciation (LLD)

A1 The value of the light source lumen depreciation (LLD) associated with the light source operating hours (replacement interval) is to be determined for the calculation of E_{Ph} , E_{PhS} , E_{Pv} and shall be obtained from data applicable to the specific luminaire and light source chosen. **A1**

For SSL luminaires, depreciation data shall be obtained from the LED chip's IESNA LM-80-15 test report and extrapolated using the IESNA TM-21-11 calculator, using the temperature of the LED chip when operating in the luminaire. *In situ* temperature measurements for the lumen depreciation of LED luminaires under normal operating conditions, as specified in AS/NZS 60598.2.3, should be made according to Annex A of IESNA LM-84-14 and the data extrapolated by the methods specified in IESNA TM-21-11, or where the exponential-fit-function procedure of IESNA TM-21-11 is not applicable, apply IEC 63013:2017. Testing shall be conducted at an ambient temperature of 40°C for Australia and 25°C for New Zealand.

The lumen depreciation caused by ageing of any optical components, e.g. optical degradation of lenses and visors should be a part of the LLD.

3.5.4 Constant light output (CLO)

CLO compensates for the light loss caused by ageing of the light source. The light source lumen depreciation value shall be determined as specified in [Clause 3.5.3](#).

As the photometric data for CLO luminaires can be provided as initial (100 % output) values or maintained values, the data needs to be verified to ensure the appropriate LLD value is assigned.

NOTE The luminaire's system power will gradually increase until the light source replacement interval is reached.

3.5.5 System (overall) design light loss factor (LLF)

The maintenance factor to be assumed for the calculation of $\bar{E}_h$, E_{Ph} and E_{Pv} , being a combination of the factors referred to in Clauses 3.5.2 and 3.5.3, shall not exceed the following:

$$LLF = LDD * LLD$$

3.6 Basis of conformance

3.6.1 **A1** Conformance of the design

The design of the lighting scheme shall be deemed to conform to this document provided—

- (a) the luminaire used conforms to AS/NZS 60598.2.3; and
- (b) when created in accordance with the relevant procedures in [Section 4](#), it satisfies the applicable light technical parameters of [Tables 3.3](#) to [3.10](#) when considered together with the applied maintenance factor and the documentation listed in [Appendix C](#) is provided. **A1**

3.6.2 **A1** Conformance of the installed lighting scheme

After installation of a lighting scheme conforming to [Clause 3.6.1](#), the installation shall be deemed to continue to conform to this document provided—

- (a) an audit of the installation shows that it accurately replicates the final design (i.e. "as built" verification);

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- (b) the final design documentation includes any changes to the original design made or agreed to by the designer in response to circumstances arising during installation and these changes do not cause non-conformance of the design with the requirements of [Clause 3.6.1](#); and
- (c) either—
 - (i) the maintenance regime referred to in [Appendix C](#), Item (k) is implemented; or
 - (ii) an alternative maintenance regime is implemented that can be demonstrated to provide a maintenance factor equal to or better than the design value

Conformance of the installed lighting scheme to the LTPs shall be based on calculation according to AS/NZS 1158.2, not by site measurement. See [Section 4](#) for the particular and relevant design methods and requirements for use in assessing conformance to the specified light technical parameters. ^(A1)

^(A1) **Table 3.3 — Values of light technical parameters for roads in local areas**

1	2	3	4
Lighting subcategory	Light technical parameters (LTP)		
	Average horizontal illuminance ^{a,b} ($\bar{E}_h$)	Point horizontal illuminance ^{a,b} (E_{Ph})	Illuminance (horizontal) uniformity ^c Cat. P (U_{E2}) (max/avg)
	lx	lx	
PR1	7	2	8
PR2	3.5	0.7	8
PR3 ^{d,e}	1.75	0.3	8
PR4 ^{d,e}	1.3	0.22	8
PR5 ^{d,e}	0.85	0.14	10
PR6 ^d	0.7	0.07	10
^a These values are maintained. ^b Conformance is achieved by being greater than or equal to the applicable table value. ^c Conformance is achieved by being less than or equal to the applicable table value. ^d See Clause 3.2 for lumen derating values for light sources with S/P ratio < 1 and/or CCT < 2 500 K. ^e When the luminaires are to be supported on existing electricity reticulation poles, the subcategories PR3, PR4 and PR5 may be reduced to the next lower subcategory PR4, PR5 and PR6 respectively. NOTE: Where there is a significant fear of crime or where required by the relevant authority then for enhanced lighting of the formed pathways, see Table 3.4. ^(A1)			

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Table 3.4 — Values of light technical parameters for pathways

1	2	3	4	5	6
Lighting sub-category	Light technical parameters				
	Average horizontal illuminance ^{a,b} ($\bar{E}_h$)	Point horizontal illuminance ^{a,b} (E_{Ph})	Illuminance (horizontal) uniformity ^c Cat. P (U_{E2}) (max/avg)	Point vertical illuminance ^{a,b} (E_{Pv})	Pathway Surround Illuminance ^{a,b,d} (E_{PhS})
	lx	lx		lx	lx
PP1	10	2	5	1	1
PP2	7	1	5	0.3	0.5
PP3	3	0.5	5	0.1	0.25
PP4 ^f	1.5	0.25	5	0.05 ^e	0.12
PP5 ^f	0.85	0.14	5	0.02 ^e	0.07
^a These values are maintained. ^b Conformance is achieved by being greater than or equal to the applicable table value. ^c Conformance is achieved by being less than or equal to the applicable table value. ^d Required unless deemed otherwise by the relevant authority (see Clause 4.6). ^e For luminaires with mounting heights of 1.5 m or less, the E_{Pv} values need not be applied. ^f See Clause 3.2 for lumen derating values for light sources with S/P ratio < 1 and/or CCT < 2 500 K. A1					

Table 3.5 — Values of light technical parameters for public activity areas (excluding car parks)

1	2	3	4	5
Lighting subcategory	Light technical parameters (LTP)			
	Average horizontal illuminance ^{a,b} ($\bar{E}_h$)	Point horizontal illuminance ^{a,b} (E_{Ph})	Illuminance (horizontal) uniformity ^c Cat. P (U_{E2}) A1 (max/avg) A1	Point vertical illuminance ^{a,b,d} (E_{Pv})
	lx	lx		lx
PA1	21	7	8	7
PA2	14	4	8	4
PA3	7	2	8	2
^a These values are maintained. ^b Conformance is achieved by being greater than or equal to the applicable table value. ^c Conformance is achieved by being less than or equal to the applicable value. ^d Where the public activity area is a lane or roadway (such as may occur in a CBD area) the light technical parameter conformance shall be determined in accordance with the rules for wide pathways as provided in AS/NZS 1158.2. A1 NOTE A1 [Text deleted.] A1				

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Table 3.6 — Values of light technical parameters for connecting elements

1	2	3	4	5
Lighting subcategory	Light technical parameters (LTP)			
	Average horizontal illuminance ^{a,b,d} ($\bar{E}_h$)	Point horizontal illuminance ^{a,b} (E_{Ph})	Illuminance (horizontal) uniformity ^c Cat. P	Point vertical illuminance ^{a,b} (E_{Pv})
	lx	lx	(U_{E2}) $\left[\frac{A_1}{A_2} \right]$ (max/avg) $\left[\frac{A_1}{A_2} \right]$	lx
PE1	35	17.5	8	17.5
PE2	Same as for highest lighting subcategory applying to areas that abut the connecting element but, where forming part of a road or pathway, to be not less than subcategory PA3 in Table 3.5 .			
PE3	Same as for highest lighting subcategory applying to areas that abut the connecting element but, where forming part of a road or pathway, to be not less than subcategory PP3 in Table 3.4.			
a These values are maintained. b Conformance is achieved by being greater than or equal to the applicable table value. c Conformance is achieved by being less than or equal to the applicable value. d For steps, the requirements assume that the noses of the treads are clearly delineated by a contrasting stripe or other equally effective means. If this does not apply, the illuminance should be at least twice the value specified. NOTE 1 It is recommended that the walls of subways be finished in a light colour to facilitate inter-reflection of light within the space. Such inter-reflected light may be taken into account in the achievement of the specified light technical parameters. $\left[\frac{A_1}{A_2} \right]$ NOTE 2 [Text deleted.] $\left[\frac{A_1}{A_2} \right]$				

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Table 3.7 — Values of light technical parameters for outdoor off-road car parks (including roof-top car parks)

1	2	3	4	5
Lighting subcategory	Light technical parameters (LTP)			
	Average horizontal illuminance ^{a,b} ($\bar{E}_h$)	Point horizontal illuminance ^{a,b} (E_{Ph})	Illuminance (horizontal) uniformity ^c Cat. P (U_{E2}) (max/avg)	Point vertical illuminance ^{a,b} (E_{Pv})
	lx	lx		lx
PC1	14	3	8	3
PC2	7	1.5	8	1
PC3	3.5	0.7	8	—
PCD ^d	—	≥ 14 and $\geq (\bar{E}_h)^d$	—	—
PCX ^e	—	≥ 5 and $\geq (\bar{E}_h)^e$	—	—

^a These values are maintained.

^b Conformance is achieved by being greater than or equal to the applicable table value.

^c Conformance is achieved by being less than or equal to the applicable table value.

^d E_{Ph} shall be determined for each PCD area in the car park and, in each case, it shall be greater than the value stated and greater than the average for the relevant/ adjacent parking module.

^e E_{Ph} shall be determined for each designated area for pedestrians to cross circulation roadways and access roadways and for each design area it shall be greater than the value stated and greater than the average for the relevant/adjacent parking module, access driveway or circulation roadway.

NOTES:

1 Conformance to the light technical parameters in [Table 3.7](#) is based on an open, unoccupied car park, i.e. free of vehicles.

2 Where raised obstructions are present, e.g. to limit vehicle movement in parking areas, these obstructions present potential hazards for pedestrians. Such obstructions should therefore be of such a material, or so finished, as to provide a high visual contrast with the paved surface.

3 See [Table 3.6](#) for the requirements that apply to connecting elements, including steps and ramps within car parks.

4 The luminaires should be positioned to highlight physical obstructions or other similar hazards to pedestrian and vehicular traffic.

5 Lighting performance requirements for indoor car parks are specified in AS/NZS 1680.2.1. [A1](#)

3.7 Control of glare and upward waste light

3.7.1 Glare

3.7.1.1 General

For the limitation of glare for luminaires used for Category P applications, two metrics are defined—

- (a) luminous intensity; and
- (b) discomfort glare.

Luminaires shall meet both criteria for the application and mounting support type used.

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3.7.1.2 **Luminous intensity**

The maximum luminous intensity for SSL and non-SSL luminaires at any azimuth angle shall not exceed the values in [Table 3.8](#). The values in [Table 3.8](#), columns 2 and 3 apply to both SSL and non-SSL luminaires. ^(A1)

Table 3.8 — Limitation of luminous intensity for SSL and non-SSL luminaires

Vertical angles ^a	Luminaire luminous flux ^c	
	≤ 4 000 lm	> 4 000 lm
Any angle from 80° to less than 90° ^b	720 cd absolute	180 cd/1 000 lm
At 90°	300 cd absolute	80 cd/1 000 lm
^a For sub-categories PR and PP, the limits apply to the luminaire in the positioned photometered (see Section 5). For sub-categories PA, PE, and PC, the limits apply to the luminaire in its designed position. ^b For sub-categories PA, PE and PC, a value of 3 500 cd absolute may be used. ^c For non-SSL luminaires, the initial lamp lumens is used. For SSL luminaires, the initial luminaire output lumens is used. ^(A1)		

3.7.1.3 **Discomfort glare from SSL luminaires**

Discomfort glare from SSL luminaires may increase due to a smaller flashed area, and a non-uniform luminance. The discomfort glare index formula is used to limit discomfort glare only from SSL luminaires as used for road and pathway lighting subcategories in Category P lighting. ^(A1)

It is given by—

$$DGI_p = I \times A_L^{-0.5}$$

The intensity 'I' is the luminous intensity, in candelas (cd), for a luminaire in the position as photometered in any azimuth direction between and including angles of elevation of 65°–85° from the downward vertical, using angles required by AS/NZS 1158.2.

The luminous area, A_L , is the sum of the orthogonally projected area/s, in square metres, in the direction of I of all the luminous parts of the luminaire, see [Appendix E](#).

The maximum DGI_p (DGI_{pmax}) is found from the set of calculated DGI_p values for all required angles of azimuth and elevation.

The DGI_{pmax} shall be no greater than the values shown in [Table 3.9](#).

An example of a typical calculation and the method to determine the luminous area is provided in [Appendix E](#).

NOTE The DGI_p is based on the method used in EN 13201-2.

It is possible that DGI_{pmax} may not coincide with the direction of I_{max} within the required range of angles.

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Table 3.9 — Limitation of discomfort glare for SSL luminaires

Recommended luminaire application	DGI _p max mounting height (H)	
	H < 6 m	6 m ≤ H ≤ 10 m
Subcategories PR and PP	35 000	40 000
LATMs, roundabouts	42 000	50 000 ^(A1)

3.7.2 Upward waste light ratio (UWLR)

For the limitation of upward waste light the maximum *UWLR* for a luminaire in the attitude in which it is installed shall be as shown in Table 3.10.

Table 3.10 — Limitation of upward waste light

Luminaire application	Maximum UWLR Light source type	
	Non-SSL	SSL
Local roads (PR and PP subcategories)	4 %	1 %
Public spaces (PA and PE subcategories)	5 %	3 %
Outdoor off-road car parks (including roof-top car parks) (PC)	Refer to AS/NZS 4282 ^(A1)	

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Section 4 Design methods and rules

4.1 Basis of design

4.1.1 General

The lighting scheme shall be designed using the methods set out in this [Clause 4.1](#), and [Clauses 4.2](#) to [4.6](#), which apply to particular road or public space elements.

Illuminance calculations shall be carried out by means of computer software to provide values of horizontal and vertical illuminance that satisfy the requirements for the relevant lighting subcategory in Tables 3.3 to 3.7. The calculation procedures shall be those specified in AS/NZS 1158.2 but may be amplified by requirements of design rules in this Section. The illuminance analysis shall exclude light that crosses property boundaries (i.e. the light may be obstructed).

NOTE [Table 4.1](#) provides a summary of the design methods for various elements. AS/NZS 1158.2 specifies the light technical parameters (LTP), calculation fields and computer methodologies to be used in applying these design methods.

For all road elements, the upward waste light ratio (*UWLR*) shall be determined in accordance with [Clause 3.7.2](#).

NOTE Luminaires should be correctly aligned with respect to the road surface so the luminaire is parallel with the grade of the road at its centre-line.

Table 4.1 — Summary of acceptable design methods

Item	Element	Design method
1	Local roads (Clause 4.2)	Illuminance-based computer calculations and design rules
2	Pathways for pedestrians or cyclists (Clause 4.6)	Illuminance-based computer calculations
3	Public activity areas (Clause 4.7)	Illuminance-based computer calculations
4	Connecting elements (Clause 4.8)	Illuminance-based computer calculations
5	Outdoor car parks (Clause 4.9)	Illuminance-based computer calculations

4.1.2 Departures from design spacing (*S*)

Calculations of illuminance in accordance with this Section will result in the determination of design spacing (*S*), which are the nominal maximums for the lighting scheme that ensure conformance to the applicable light technical parameters. However, in practice, features such as intersections, driveways, trees or power distribution poles may prevent the location of lighting poles/columns in positions that will enable precise achievement of *S*.

Where necessitated at isolated points in the installation, e.g. due to the fixed position of an existing power distribution pole or to a physical obstruction preventing a lighting pole being sited in the desired location, the spacing in individual spans may exceed *S* by not more than 10 %, provided that such non-conforming spacing does not occur for more than one span and does not reoccur for the next three consecutive spans.

4.2 Local roads

4.2.1 Straight sections

For straight sections of road designated subcategories PR1 to PR6, conformance to the values of $\bar{E}_h$, E_{ph} and U_{E2} in Table 3.3 shall be assessed using the methods set out in [Clause 4.1.1](#).

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Where the road incorporates a designated, formed pathway that justifies the provision of a higher lighting level subcategory (i.e. PP1, PP2, PP3, PP4 or PP5), only the pathway and not the whole of the road reserve shall be lit to the applicable subcategory in accordance with [Clause 4.1.1](#).

4.2.2 Curved sections of road and bends

For curved sections of road and bends, conformance to the specified values for $\bar{E}_h$, E_{Ph} and U_{E2} shall be deemed to have been achieved as follows:

- (a) The luminaire spacing shall not exceed that required for a straight section of the same road reserve width.
- (b) A straight line joining successive luminaires shall lie within the road reserve.
-  (c) Where the orientation of the $C = 90^\circ/270^\circ$ plane of the luminaire is $\geq 15^\circ$ from an adjacent luminaire, an illuminance analysis shall be carried out over the extent of the curve to ensure that the requirements for $\bar{E}_h$, E_{Ph} and U_{E2} in [Table 3.3](#) are met. The illuminance analysis shall exclude light that crosses property boundaries (i.e. the light may be obstructed), particularly at intersections, curves and bends. The extent of the calculation field shall be over the full extent of the curve or bend. Separate calculations shall be undertaken between each span. The grid spacing used in both directions for the curve section of road shall not be greater than 1 m. 

NOTE The extent of the calculation field can be over the full extent of the curve or by separate fields between each span.

4.3 Intersections

4.3.1 Roads with Category P lighting intersecting with roads with Category P lighting

For intersections of roads provided with Category P lighting, the following design rules shall apply:

- (a) A luminaire or luminaires shall be located within the intersection within the hatched area (see [Figure 4.1](#)).
- (b) The luminaire spacing for the through roads, across the intersection, shall be not more than that determined in accordance with [Clause 4.2.2](#).
- (c) Where the through roads are provided with, or justify the provision of, different levels of Category P lighting, the luminaire installed within the shaded area shall be of a type that satisfies the requirements applicable to the higher of the lighting subcategories.
-  (d) Where the orientation of the $C = 90^\circ/270^\circ$ plane of the luminaire is $\geq 15^\circ$ from adjacent luminaire(s) (see [Figure 4.1](#)) on the same roadway, an illuminance analysis shall be carried out over the extent of the intersection as shown by the hatched area in [Figure 4.1](#) to ensure that the requirements of $\bar{E}_h$, E_{Ph} and U_{E2} of [Table 3.3](#) are met. The illuminance analysis shall exclude light that crosses property boundaries (i.e. the light may be obstructed). 
- (e) For T-intersections, see [Figure 4.2](#). One of the following design rules shall apply:
 -  (i) Where the location of the luminaire in the through road is opposite the joining road and within the hatched area, the first luminaire in the joining road shall be located a distance $\leq 0.6S$ from the luminaire in the through road. 
 - (ii) Where the luminaire required for the T-intersection is not associated with the through road, but forming part of the lighting scheme for the joining road, then this luminaire shall be located within the hatched area.

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- (iii) Where the location of the luminaire in the through road within the hatched area is not opposite the joining road [see [Figure 4.2\(iii\)](#)], the first luminaire in the joining road shall be located a distance from the through road property line of no more than $0.25S$ for the joining road. Line of sight is not necessary in this case.
- A1** (iv) Any other alternative layout may be used provided calculations demonstrate conformance with $\bar{E}_h$, E_{Ph} and U_{E2} in [Table 3.3](#) are met. The illuminance analysis shall exclude light that crosses property boundaries (i.e. the light may be obstructed), particularly at intersections, curves and bends. **A1**

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A1 Dimensions in metres

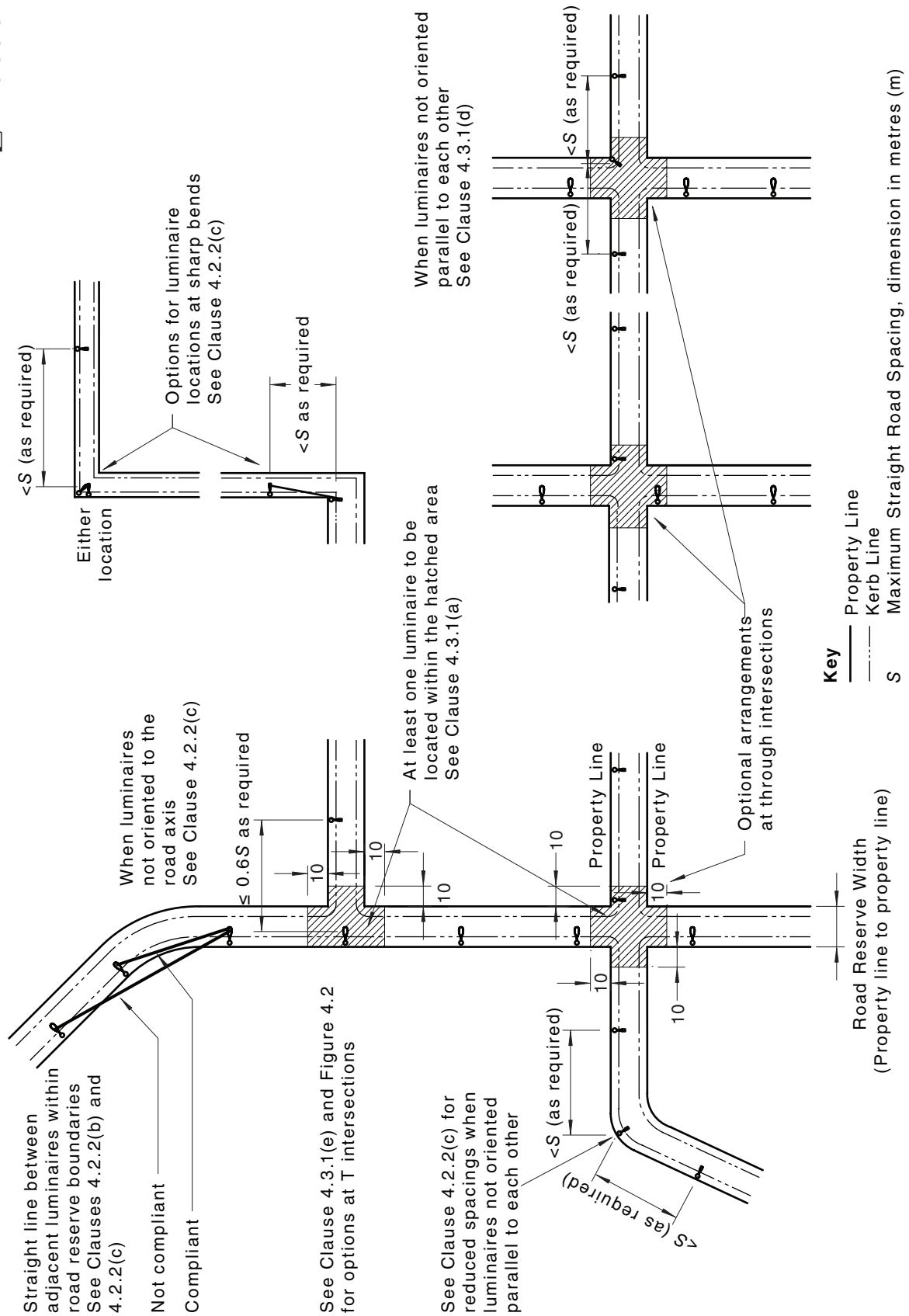


Figure 4.1 — Requirements for locations of luminaires on curves, sharp bends and intersections **A1**

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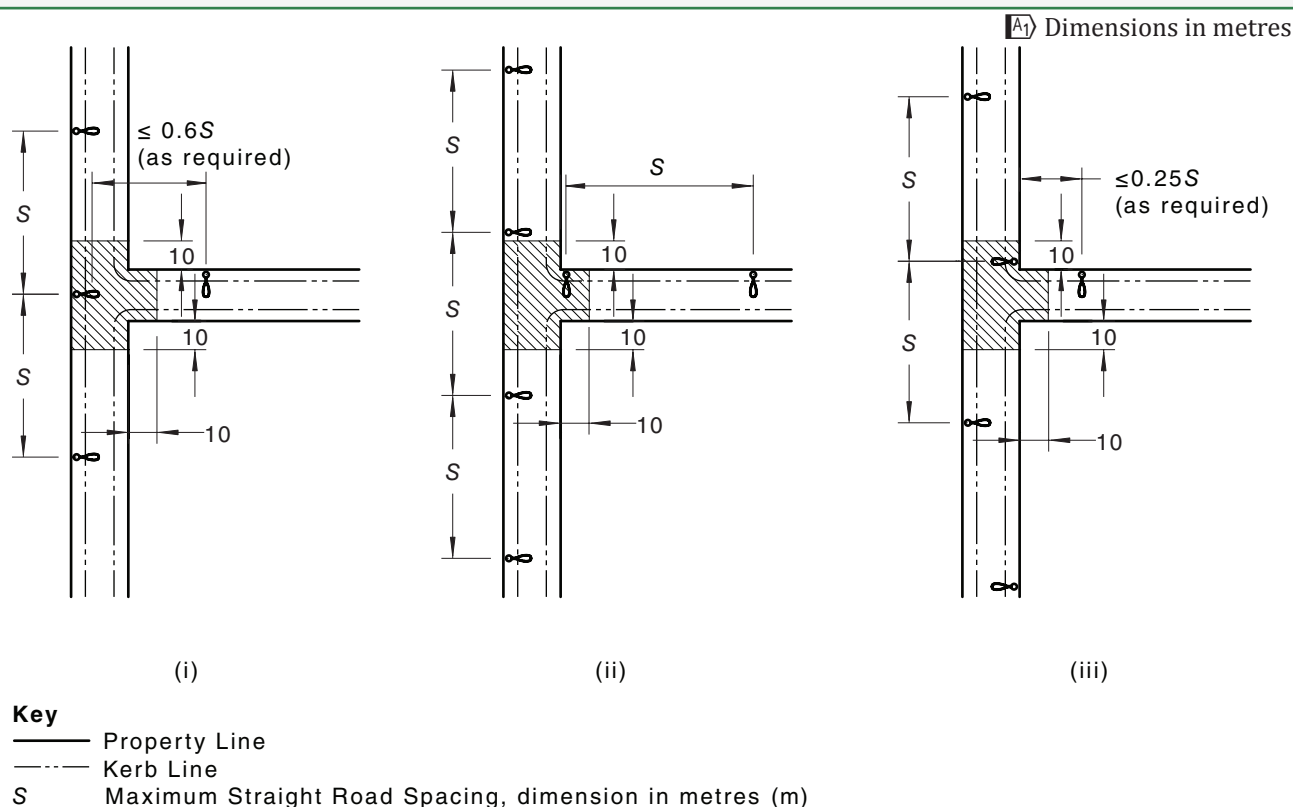


Figure 4.2 — Requirements for locations of luminaires at T-intersections [A1]

4.3.2 Roads with Category P lighting intersecting roads with Category V lighting

Where a road that is to be provided with Category P lighting intersects a road with Category V lighting, or a road for which Category V lighting is appropriate, the requirement for the Category V lighting is given in AS/NZS 1158.1.1. On the road provided with Category P lighting, the design rule shall be as follows:

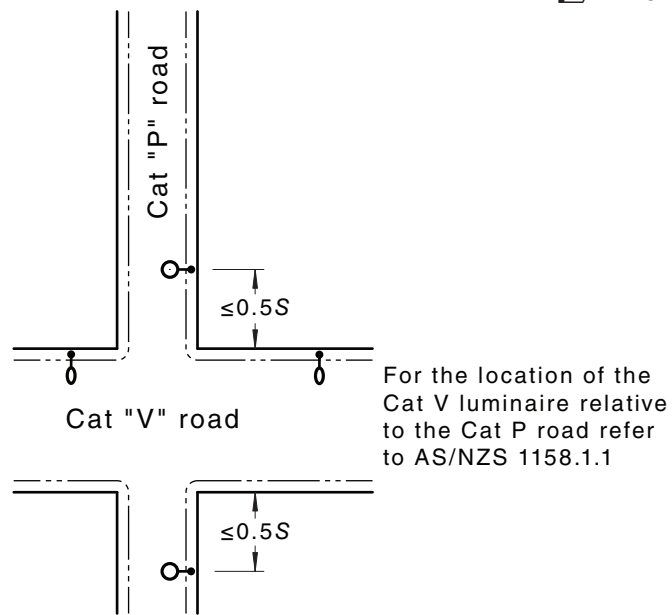
- (a) *Where no channelization exists on the minor road* The Category P luminaire shall be located at a distance of not more than 50 % of the spacing for the road determined for conformance to [Clause 4.2.1](#), from the limits of the intersection defined by the prolongation of the property lines. See [Figure 4.3\(a\)](#).

- [A1] (b) *Where the minor road contains channelization that creates a traffic separation commencing within 10 m of the intersection* The minimum illuminance of the nose of the channelization shall be 3.5 lx in the Category P road. Those parts of the islands abutting the Category V road shall be lit to the appropriate Category V level, see [Figure 4.3\(b\)\(i\)](#) for the design area. Where the channelisation extends more than 10 m into the Category P road, see [Figure 4.3\(b\)\(ii\)](#). [A1]

The first luminaire of the Category P lighting scheme shall be at a distance from the last luminaire of the Category V lighting scheme not greater than that determined for conformance to [Clause 4.2.1](#).

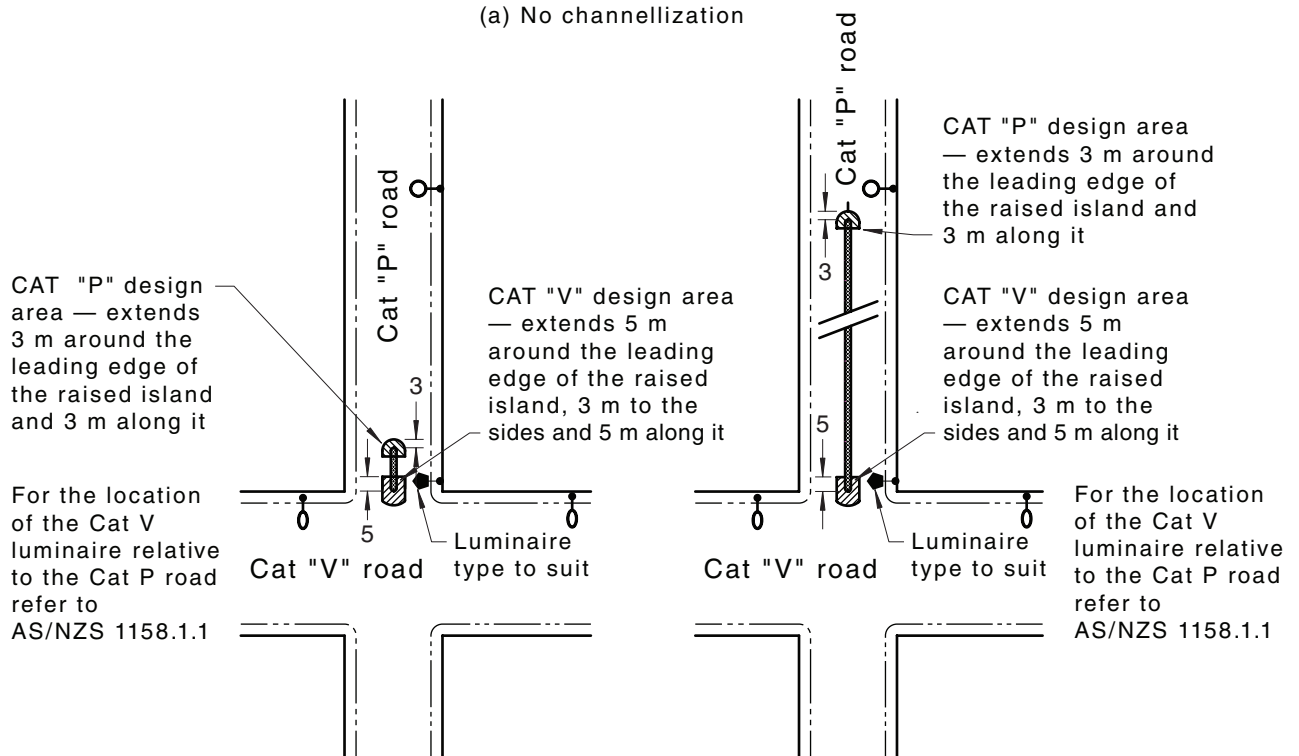
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 Dimensions in metres



See Clause 4.3.2 (a)

(a) No channellization



See Clause 4.3.2 (b)

(b)(i) With channellisation ≤ 10 m

See Clause 4.3.2 (b)

(b)(ii) With channellisation > 10 m

Key

— Property Line

--- Kerb Line

S Maximum Straight Road Spacing, dimension in metres (m)

Figure 4.3 — Requirements for luminaires and design areas where roads lit to category P intersect with roads lit to category V 

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4.3.3 Cul-de-sacs

4.3.3.1 General

Illumination of any shape of cul-de-sac shall be determined by ensuring a luminaire layout that provides the whole of the public road reserve with an illumination level that meets the specified E_{ph} requirement for the subcategory selected as per Column 3 of Table 3.3. Luminaire types shall be of a similar type and wattage as those illuminating the entrance streets to the cul-de-sac—although variations in the optics and installed tilt can be used provided the requirements of Tables 3.8 (Glare) and 3.9 (*UWLR*) are met. Luminaires layouts should preferably be at the same height and set back as those in the adjacent feeder road.

Luminaire locations shall maintain line-of-sight requirements as specified in [Clause 4.2.2\(b\)](#).

4.4 Pedestrian refuges

Where pedestrian refuges are located on roads requiring Category P lighting, the maintained horizontal illuminance over the surface of the refuge, within the design area shown in [Figure 4.4](#), shall be not less than 3.5 lx.

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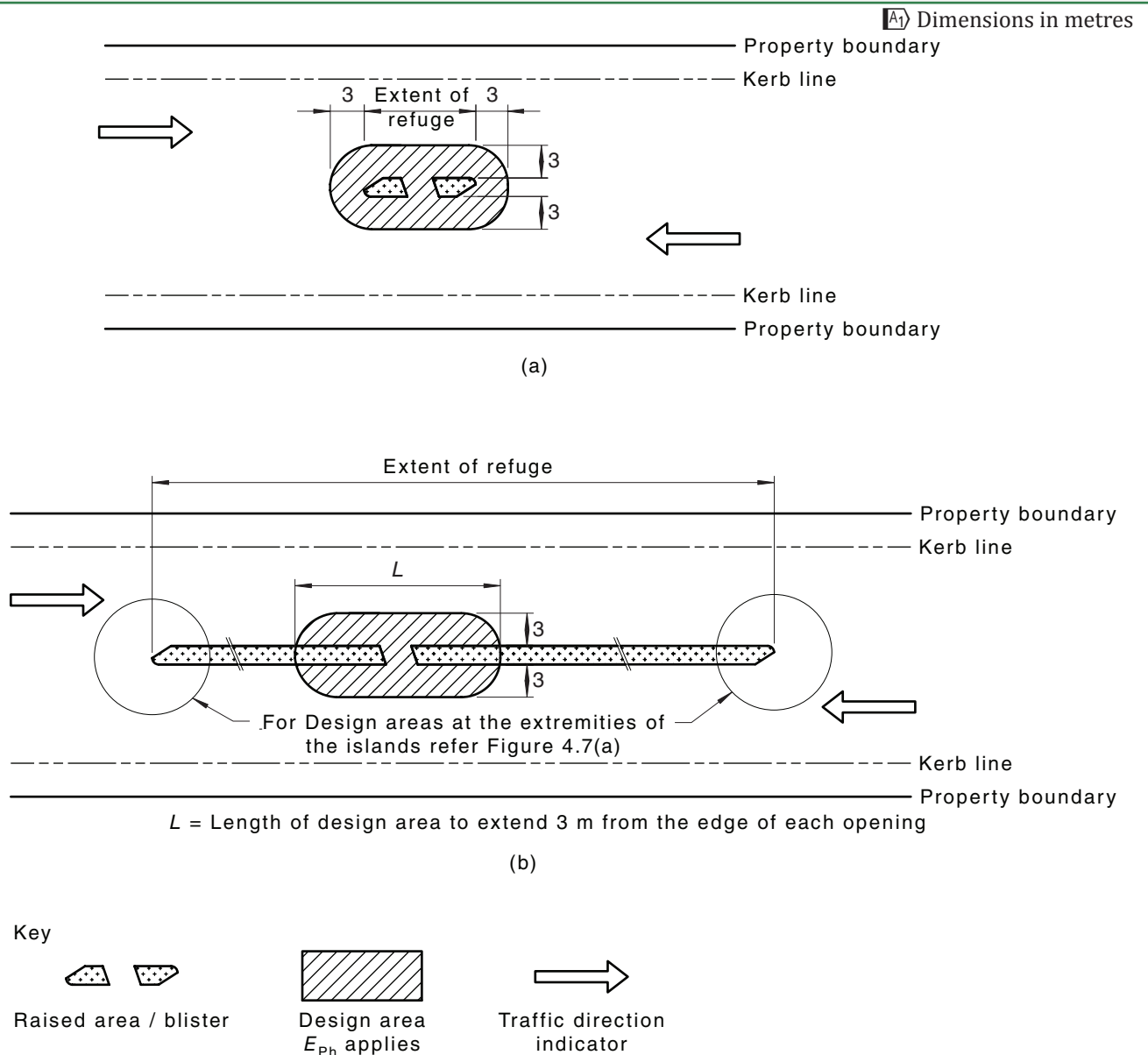


Figure 4.4 — Minimum illuminance calculation design area for pedestrian refuges [A1]

4.5 Local area and traffic management devices

4.5.1 General

Where a discrete traffic management device (e.g. roundabout, speed-hump, slow point or chicane) is installed on the carriageway as part of a local area traffic management system, and would present a potential hazard to a night-time driver, the potential hazard shall be identified to approaching drivers by conformance to [Clause 4.5.2](#) or [Clause 4.5.3](#) as applicable.

Because all traffic management devices are intended to have one of two possible effects on the traffic, depending on the purpose of the particular road, [Clauses 4.5.2](#) and [4.5.3](#) detail two significantly different lighting approaches as follows:

- (a) The intention to regulate traffic ([Clause 4.5.2](#)).
- (b) The intention to deter traffic ([Clause 4.5.3](#)).

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4.5.2 Devices intended to regulate traffic

4.5.2.1 General

Where the aim is to slow traffic and regulate its flow at conflict points but not to deter or reduce the volume of through traffic, the intended effect of a traffic management device is to cause the driver to reduce speed and maintain a reduced speed while travelling through the device. The intention of the lighting associated with the device is to alert the driver of both the presence and nature of the device so that the driver can adjust speed accordingly.

To achieve this, an illumination of not less than 3.5 lx over the relevant design area shall be provided.

4.5.2.2 Trafficable roundabout

A trafficable roundabout is where the channelization consists of a small painted and/or raised trafficable island.

The extent of this small painted and/or raised trafficable roundabout shall be taken as the design area.

NOTE 1 As a guide, illumination of the roundabout may be achieved using two luminaires consistent with the type, wattage and mounting height as used on the approaching roadways, and positioned one on each of the approach roadways, with priority to the placement being influenced by the greater volume of traffic and principal traffic flow associated with the respective approaches, in accordance with [Figure 4.5](#).

NOTE 2 If necessary, additional luminaires may need to be used on the other feeder roads.

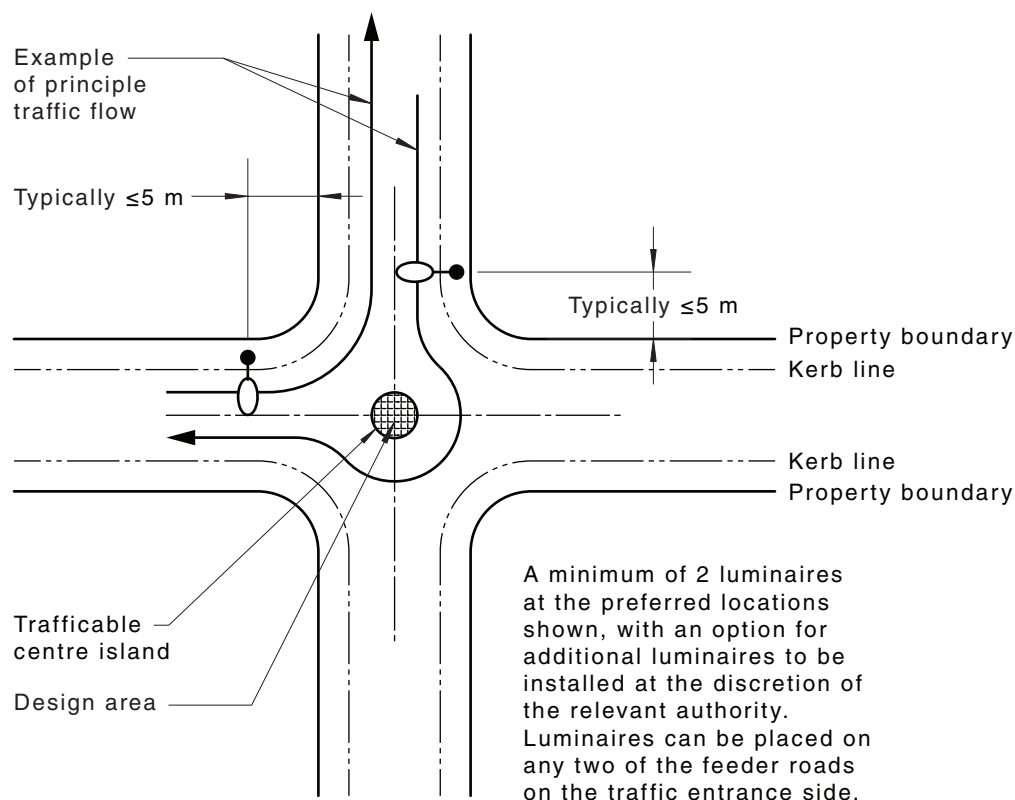


Figure 4.5 — Minimum illuminance calculation design area trafficable roundabout

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4.5.2.3 Complex channelized roundabout

A complex channelized roundabout is where channelization may consist of both a non-trafficable centre island plus any type or combination of splitter islands, refuges or centre-medians that in some way define the passage of vehicular traffic flow through the device.

An arrangement of luminaires shall be established to provide $E_{ph(min)}$ values of not less than 3.5 lx over the design area as defined in Figure 4.6, and shown to extend out to 3.0 m beyond the outer radius of the circulating carriageway and square to the carriageways of the approaching roads as shown, and including any further design area applicable to the components of the associated channelization, i.e. the design area shall include any part or portion of a raised island, refuge or median that may fall outside this primary design area and as covered separately under the lighting rules that apply to such traffic channelization devices.

Dimensions in metres

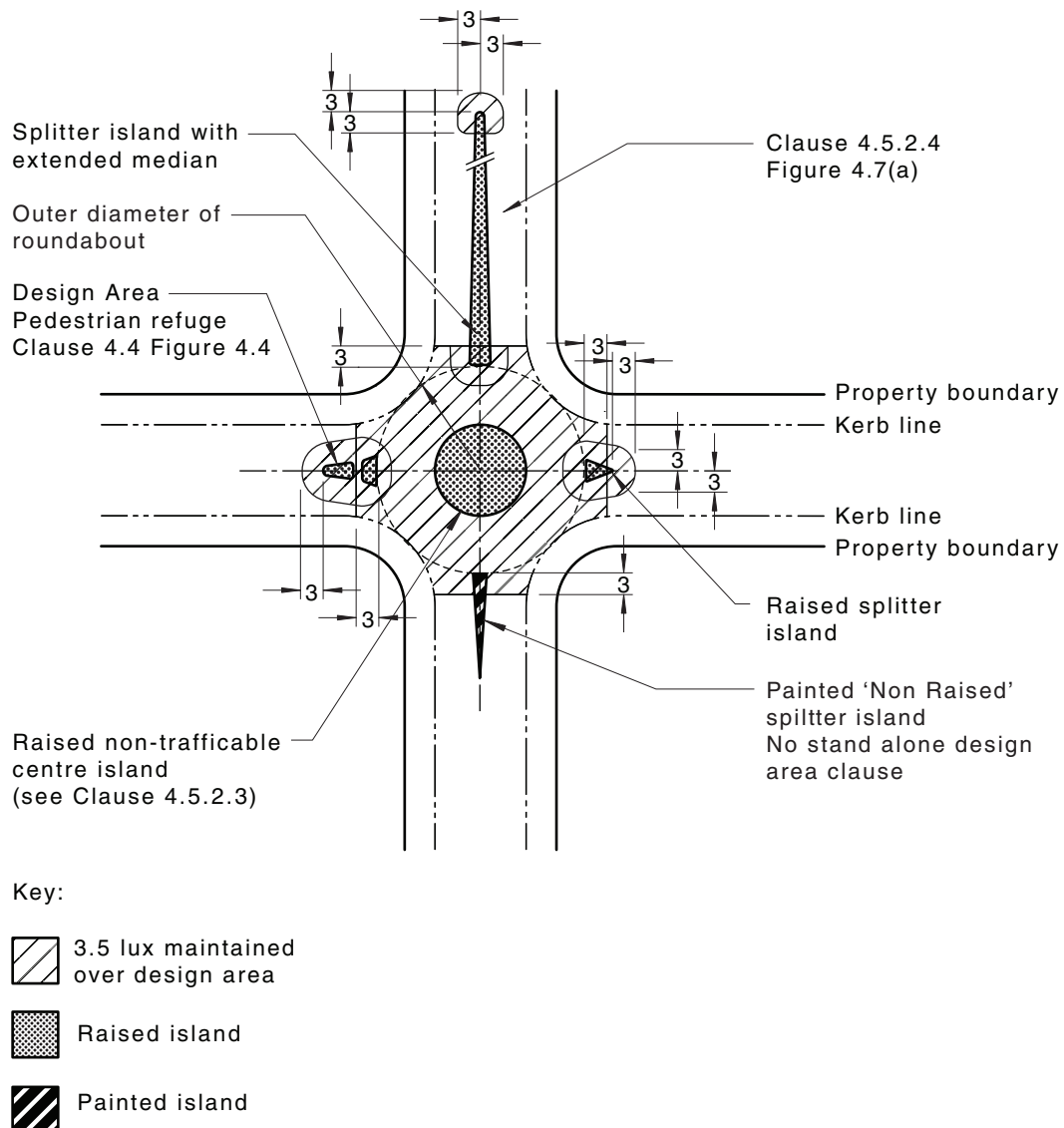


Figure 4.6 — Minimum illuminance calculation design area for complex channelized roundabout

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4.5.2.4 Other local area traffic management (LATM) devices

For other LATM devices, see [Figure 4.7](#) for an illustration of the principles for determining the minimum design area for illuminance calculations associated with typical LATM devices other than roundabouts.

Unlike the entrance to a traffic management device, additional illumination of the exit from the device is not required as vehicles approach this at a sufficiently low speed due to the device itself. However, in most cases the exit is immediately adjacent to the entrance for traffic in the reverse direction.

[A₁] Where a vertical deflection device such as a raised safety platform, speed hump/table or pedestrian platform is installed in a Category P road within 10 m of an intersection with a Category V road, the design area shall be in accordance with [Figure 4.7\(b\)](#), with the Category V side performance in accordance with Part 1.1, Table 2.2, Columns 8 and 9 and the Category P side to achieve a minimum of 3.5 lx. **[A₁]**

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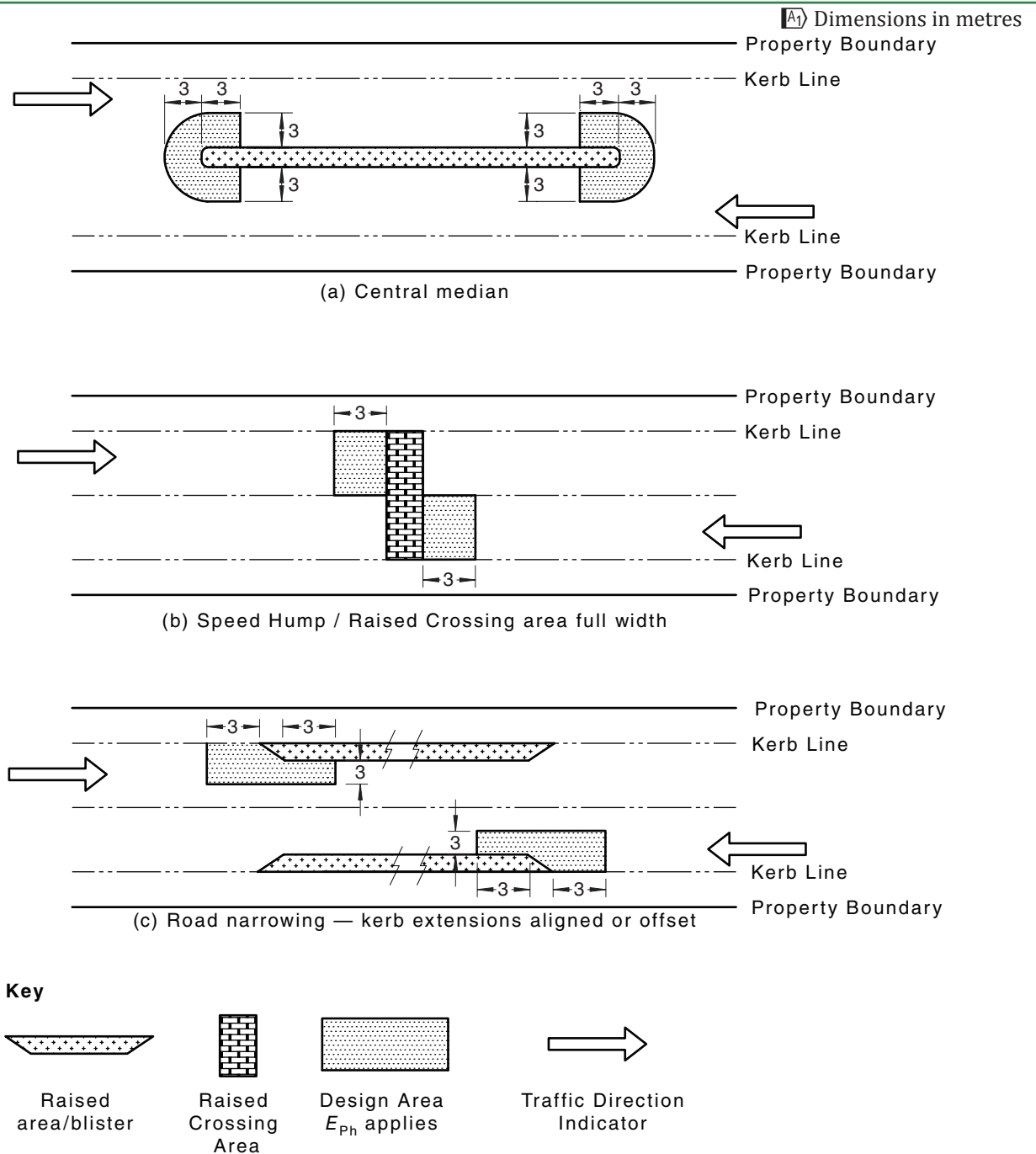


Figure 4.7 — Illustration of the principles for determining the minimum design area for illuminance calculations for traffic management devices other than roundabouts [A1]

4.5.3 Devices intended to deter traffic

When the purpose of the road is such that the intention of traffic management devices is generally to slow and deter traffic other than vehicles with their origin or destination in that road, the devices shall be identified to drivers using one or both of the following methods:

(a) Using reflectors:

The traffic management device shall be fitted with adequate reflective devices in accordance with—

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- (i) in Australia: AS 1742.13; or
- (ii) in New Zealand: *Traffic Control Devices Manual*,
so as to unambiguously demark it to oncoming motorists.
- (b) A luminaire forming part of the general lighting scheme for the road shall be located, ideally in close proximity to the traffic control device, and in no case more than $0.25S$ from each entry point.

4.6 Pathways for pedestrians or cyclists

4.6.1 A1 General

Where the pathway does not form part of a road reserve, subcategory PP1, PP2, PP3, PP4 or PP5 is applicable.

Where the pathway forms part of a local road that is to be lit to subcategories PR1 to PR6, no special lighting requirements apply other than those specified in [Clause 4.2](#) for the whole of the road reserve.

Where a pathway within a road reserve justifies, or is to be provided with a pathway subcategory, conformance to the values of $\bar{E}_h$, E_{Ph} and U_{E2} and E_{Pv} as applicable in [Table 3.4](#) with the exemption of the surround (path) illuminance per Footnote d, shall be assessed using the methods set out in [Clause 4.1](#).

At the start/end of the pathway a luminaire should be placed no greater than $0.33 \times S$ (the maximum possible for a straight section) from the start/end. A1

4.6.2 A1 Pathway surround illuminance

Unless deemed otherwise by the relevant authority the minimum pathway surround point horizontal illuminance shall also be provided over two calculation fields, one either side of the pathway. The width of each field shall be 5 m where a suitable verge exists, otherwise the field may be taken up to any structure/fence/property boundary from the edge of the pathway. Where a residential property is located within 3 m of the pathway, the surround (path) point horizontal illuminance may not be provided on that side.

Calculation points for the pathway surround point horizontal illuminance shall be located per AS/NZS 1158.2 for pathways greater than 2 m in width. The overall width of the field is a combination of both surround fields and the pathway, e.g. 12 m for a 2 m wide pathway.

The outermost points shall not be more than half a grid spacing from the edge of the surround fields. A1

4.6.3 A1 Curved sections

For curved sections of pathway an illuminance analysis shall be carried out over the extent of the curve to ensure that the requirements for $\bar{E}_h$, E_{Ph} , E_{Pv} , U_{E2} , and the pathway surround point horizontal illuminance in [Table 3.4](#) are met. Alternatively, conformance shall be deemed to have been achieved when—

- (a) the luminaire spacing shall not exceed $0.8 \times S$ (the maximum possible for a straight section of the same path width);
- (b) a straight line joining successive luminaires shall lie within a boundary that is 2 m either side of the path width; and
- (c) where the orientation of the $C = 90^\circ/270^\circ$ plane of the luminaire is $\geq 15^\circ$ from an adjacent luminaire an illuminance analysis shall be carried out over the extent of the curve to ensure that the requirements for $2 \times \bar{E}_h$, $2 \times E_{Ph}$, and U_{E2} in [Table 3.4](#) are met.

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The illuminance analysis shall exclude light that crosses property boundaries (i.e. the light may be obstructed), particularly at intersections, curves and bends. The extent of the calculation field shall be over the full extent of the curve. Separate calculations shall be undertaken for each affected span. ^(A1)

4.7 Public activity spaces

Conformance to the values of $\bar{E}_h$, E_{Ph} and U_{E2} and E_{Pv} in [Table 3.5](#) shall be assessed using the methods set out in [Clause 4.1](#).

4.8 Connecting elements

4.8.1 Stairways

Conformance to the values of $\bar{E}_h$, E_{Ph} and U_{E2} and E_{Pv} in [Table 3.6](#) shall be assessed using the methods set out in [Clause 4.1](#).

NOTE Requirements and recommendations for the lighting of indoor stairways are given in AS/NZS 1680.0 and AS/NZS 1680.2.1 respectively.

4.8.2 Ramps, footbridges (excepting stairs), subways and pedestrian ways

Conformance to the values of $\bar{E}_h$, E_{Ph} and U_{E2} and E_{Pv} in [Table 3.6](#) shall be assessed in the same manner as for pathways using the methods set out in [Clause 4.1](#).

4.9 Car parks

4.9.1 General

^(A1) NOTE For the purpose of this Clause, some terms are used that have been defined in AS/NZS 2890.1. For convenience, these definitions are reproduced in [Clause 4.9.2](#). ^(A1)

Conformance to the values of $\bar{E}_h$, E_{Ph} and U_{E2} and E_{Pv} , as applicable, in [Table 3.7](#) shall be assessed using the methods set out in [Clause 4.1](#).

The assessment shall take into account the various elements (see [Figure 4.8](#)) which shall be treated as follows:

- (a) Design calculation areas containing parking spaces, parking aisles and areas where reversing vehicles and pedestrians are both expected (typically anywhere within 5 m of the mouth of a parking space) shall be in accordance with all of the required LTP values of the relevant subcategory in [Table 3.7](#). Areas of a car park containing these elements are generally referred to as 'parking modules' and, where two or more parking modules are adjacent to each other, they shall be treated as one for the purpose of defining the relevant design calculation area.

For areas of a car park where the calculation design area is narrower than 15 m values of E_{Pv} shall be calculated for a line of points at not greater than 5 m spacing located along the centre-line of the applicable design calculation area, with the end points not greater than 7.5 m from the extremities of that area.
- (b) Sections of roadway that access parking aisles but do not directly access parking spaces (i.e. those areas marked 'circulation roadway' and 'ramp' in [Figure 4.8](#)) shall meet the requirements of $\bar{E}_h$, E_{Ph} and U_{E2} for subcategory PC1, PC2 or PC3 as appropriate for the overall car park. E_{Pv} is not required.
- (c) Access driveways and similar long lengths of road (access roadways), such as those that link car parks or which adjoin or dissect a car park but which are neither parking aisles nor circulation aisles or roadways, shall be illuminated to not less than 50 % of each of the values

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$\bar{E}_h$ and E_{Ph} that are required for whichever of the subcategories PC1, PC2 or PC3 is appropriate for the adjacent car park. E_{Pv} is not required.

- (d) Designated disabled parking spaces shall meet requirements for subcategory PCD.
- A1** (e) Designated areas for pedestrians to cross circulation roadways and access roadways (excluding access driveways) shall meet the requirements of subcategory PCX. Lighting subcategory PCX does not apply to designated areas located in parking aisles. **A1**

4.9.2 Definitions of terms in Clause 4.9.1 from AS/NZS 2890.1

4.9.2.1 Access driveway

A roadway extending from the edge of the frontage roadway to the property boundary to connect with the first ramp, circulation roadway, parking aisle or domestic driveway encountered, and carrying one, or two-way traffic.

4.9.2.2 Circulation roadway

A roadway within an off-street car park which is used solely for circulation and to gain access to parking aisles, and on which there is no parking.

4.9.2.3 Parking aisle

A roadway or an area of pavement used by vehicles to gain access to, and to manoeuvre into and out of parking spaces.

4.9.2.4 Parking module

A parking aisle together with a single row of parking spaces on one or both sides, but excluding ramps or circulation roadways which take off within the module.

4.9.2.5 Ramp

A circulation roadway which connects an access driveway to an off-street car park on a substantially different level, or which connects two levels in a multi-level car park.

A1

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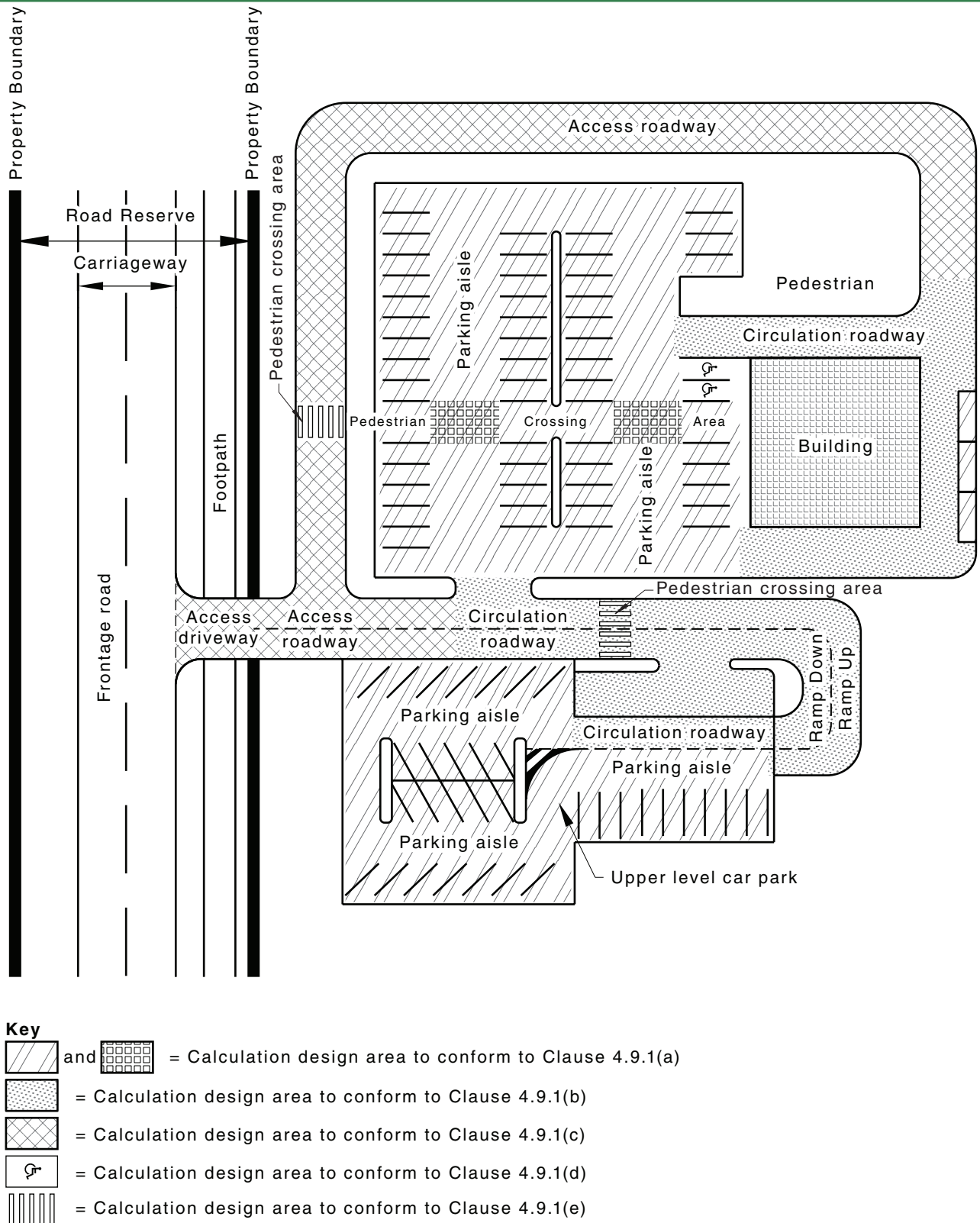


Figure 4.8 — Terminology and calculation zones for off-road car parking ^(A1)

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4.9.2.6 Pedestrian crossing area

A designated area which is a preferred location for pedestrians to cross access driveways, circulation roadways or parking aisles. These areas might be unmarked, have "Zebra" style markings or otherwise. 

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Section 5 Provision of luminaire data

All photometric data submitted in support of a design shall be accompanied by a statement on the source of the data and shall be substantiated by reference to test reports from testing performed in accordance with CIE S 025/E or IESNA LM-79-19 by laboratories that meet the requirements of AS ISO/IEC 17025.

NOTE Accreditation bodies which are signatories to the International Laboratory Accreditation Cooperation (ILAC) Mutual Recognition Arrangement (MRA) for testing laboratories may be able to offer accreditation against the requirements of AS ISO/IEC 17025. A listing of ILAC signatories is available from the ILAC website (www.ilac.org). In Australia and New Zealand, the National Association of Testing Authorities (NATA) and International Accreditation New Zealand (IANZ) are signatories to the ILAC MRA.

The following luminaire data shall be obtained to facilitate the computer calculations required for the design of Category P lighting in accordance with [Section 3](#) and the derivation of any relevant design aids:

- (a) A luminous intensity distribution table (i.e. an I-table) for the luminaire which encompasses the photometric array as detailed in Table 4.1 of AS/NZS 1158.2:2005. The I-table shall incorporate a unique identification number and the date of issue.
- (b) The photometric test procedures shall be as specified in CIE S 025/E or IESNA LM-79-19. The luminaire shall be mounted in the photometer with the axis of the fixing spigot entry horizontal (at 0° tilt) for side-entry luminaires and vertical for both post-top (i.e. bottom entry) and top-entry luminaires.
- (c) For non-SSL luminaires:
 - (i) The photometric data shall be accompanied by a statement of the type, nominal wattage, relevant dimensions, shape and envelope finish of the light sources used in the luminaire for which the data are quoted.
 - (ii) The light source operating parameters, i.e. voltage, current and wattage, both in the luminaire during testing and in free air shall be provided.
 - (iii) Any special characteristics affecting photometric performance shall be stated.
- (d) For SSL luminaires:
 - (i) The reporting of data shall be as per CIE S 025/E or IESNA LM-79-19.
 - (ii) The photometric data shall be accompanied by a statement that includes all the requirements of the report section of IESNA LM-79-19 or CIE S 025/E.
 - (iii) The LED forward current.
 - (iv) The optical dimensions of the luminous area as defined in [Appendix E](#) shall be provided.
 - (v) The maximum DGI_p .

Appendix A (informative)

Selection of lighting subcategory based on fear of crime or need to enhance amenity

A.1 Introduction

This Appendix provides information to assist in applying criteria adopted in [Tables 2.1](#) to [2.3](#), and [2.5](#) for the selection of an appropriate lighting subcategory based on—

- (a) the need to lessen the fear of crime; and
- (b) the need to enhance the amenity of the locality.

A.2 Fear of crime²⁾

This criterion has been introduced so that a level of lighting may be selected in order to reduce the fear of crime, to promote a feeling of safety.

If a community has a fear or heightened perception of crime occurrence (which may or may not be justified) it will lead to a general fear of personal involvement. This reduces the quality of life for individuals and also leads to changes in lifestyle, including not going out at night.

Several studies (Boyce et. al. 2000, Fotios, Unwin et. al. 2015, Loewen et. al. 1993, Fotios, Liachenko-Monteiro et. al. 2017, Fotios and Gibbons 2018, Fotios, Liachenko-Monteiro et. al. 2018) have demonstrated that road lighting in streets and public spaces engenders a feeling of increased safety. These studies show that illumination and good uniformity creates an increase in the perceived level of safety, leading to greater use of the night-time environment.

Boyce et. al. 2000 found that an illuminance of 30 lux for car parks was near optimal, for pedestrian footpaths Fotios and Liachenko-Monteiro 2017 found levels of 7-10 lux were acceptable. Fotios and Liachenko-Monteiro et. al. 2018 looked at the perception of safety in an urban environment as a function of the mean illuminance, illuminance uniformity and minimum illuminance. They found that minimum illuminance was the best predictor of perceived safety and that a minimum horizontal illuminance of 2 lux was sufficient.

These levels show good alignment with the levels of illuminance specified in [Section 3](#).

A.3 Need to enhance amenity

This criterion has been introduced so that a level of lighting may be selected where a higher degree of amenity is desired, e.g. in a new housing development or in a refurbishment of a civic area. The lighting will often be one of a number of items being implemented to enhance amenity.

A.4 Example of procedure

The following example illustrates the procedure to be followed in applying the selection criteria to obtain the applicable lighting subcategory. The example assumes a formed path through a small park,

2) AS/NZS ISO 31000 and SA/SNZ HB 436 provide information on the methodologies of risk assessment.

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which offers a short cut between a railway station and a housing development in a suburb; it is proposed to light the path. To determine the applicable lighting subcategory, the following steps are taken:

- (a) Select the appropriate table in [Section 2](#). In this case it will be Table 2.2 based on the following:
 - (i) *General description* Park paths (Column 1).
 - (ii) *Basic operating characteristic* Pedestrian/cycle only (Column 2).
- (b) Evaluate the selection criteria separately (Columns 3 and 4) as follows:
 - (i) *Activity—medium* High density (cluster housing) development generating commuter traffic accessing the station by foot due to its close proximity.
 - (ii) *Fear of crime—medium* Generally low fear of crime in suburban area but is adjacent to station and is potential crime spot, plus the isolated and unlit path deters use.
- (c) Select the applicable lighting subcategory (Column 5) as follows:
 - (i) *Activity—medium* Subcategory PP2 or PP4.
 - (ii) *Fear of crime—medium* Subcategory PP2.

The applicable lighting subcategory is the highest placed one in the table and, in this case, is subcategory PP2 as determined by the assessment of fear of crime and activity.
- (d) Determine the required lighting values from the appropriate table in [Section 3](#), in this case Table 3.4. Use lighting subcategory PP2 in Column 1 with the prescribed LTPs in Columns 2, 3, 4 and 5.

The example shows the need for much initial information from the client (e.g. the developers and through them, the possible stakeholders—community, local government, rail authority and police) and a continuing client/designer interaction in order to achieve the client's objectives but also to understand the implications in terms of the necessary lighting level involved.

Appendix B **(informative)**

Design brief

B.1 General

This Appendix sets out the minimum technical information that should be provided to the designer of a Category P lighting scheme. However, information regarding various other aspects specific to the particular project might be required to make up a complete design brief. Note that the design requirements will differ considerably between a Category P road and a public space, such as a pedestrian mall.

B.2 Site plans

The client (usually the applicable local government authority) will often provide site plans for the project however for some projects the client may request the designer gather the site information and develop their own site plans. Site plans should include the following:

- (a) Relevant features, e.g. intersections, local area traffic management (LATM) devices, medians, trees, drainage, lengths, pathways and parking spaces.
- (b) Relevant dimensions, especially of elements to be lit, e.g. road reserve width, path widths, dimensions of public spaces.
- (c) Any proposed changes to the existing road geometry.
- (d) Cross-sections, e.g. lanes and footpaths property boundaries.
- (e) Electricity authority details, e.g. poles, overhead power supplies and existing lighting.
- (f) Relevant underground ducting or services.
- (g) Clear start/cut-off points where plans overlap.
- (h) Extent of the lighting required, including nominated distances.

B.3 Lighting scheme performance requirements

Design brief should include the following:

- (a) Performance requirements of the lighting scheme.
- (b) The subcategory assigned to each road or public space element in the lighting scheme.
- (c) For lighting schemes that use CMS controls for adaptive lighting application, the design brief should designate if one or more lower level subcategories are applicable at off-peak times.
- (d) The subcategories assigned to abutting streets and relevant land areas.

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B.4 Specific requirements

Where applicable, the design brief should also include:

- (a) Any special equipment requirements, e.g. preferred luminaries, mounting heights, outreaches, column types and arrangements.
- (b) Relevant local government authority policies on equipment and pole placement.
- (c) Any regulatory limitations, e.g. civil aviation, marine and railway.
- (d) Clear indication of areas where lighting structures are not permissible.
- (e) Identification of areas of civic, historical, or special significance, including observatories.
- (f) Proposed/future road improvements that may need to be considered.
- (g) The provision of power density indicator (PDI) and annual energy consumption indicator (AECI) calculated values for the lighting scheme.

Appendix C (normative)

Documentation required for demonstrating conformance to this standard

The following documentation shall be prepared in order to demonstrate conformance to the requirements of this Standard (see [Clause 3.6](#)):

- (a) A statement shall be prepared by a competent person that either—
 - (i) states that the design meets both the design brief and all requirements of this Standard and shall include details of any conforming increases in maximum spacing in accordance with [Clause 4.1.2](#) that have been utilized in the design process; or
 - (ii) identifies and justifies any aspects of the design that do not conform with either the design brief, or this Standard, or both.
- (b) The essential details of all elements of the road or public space to be lit.
- (c) The selected lighting subcategory and the off-peak lighting subcategories in an adaptive lighting scheme.
- (d) Details of the lighting arrangement and geometry, e.g. spacing, mounting height, overhang, tilt angle.
- (e) Details of the lighting columns, e.g. location (including offset), type, height, material and finish.
- (f) Details of the luminaires and light sources to be used, e.g. luminaire identification details, light source type, rating and luminous flux.
- (g) A scaled plan showing significant road or public space features (e.g. kerbs, property boundaries) and the proposed lighting scheme (e.g. pole locations).
- (h) For each of the road elements or public spaces involved, the design method used and the values of the light technical parameters obtained, compared to the limiting values given in Tables 3.3 to 3.7.
- (i) The origin of the photometric data for the luminaires and light sources.
- (j) The name and source of the computer program used and a statement of its conformance to AS/NZS 1158.2.
- (k) The maintenance factor used in the design and the basis for that maintenance factor, including the following:
 - (i) Luminaire ingress protection rating specified.
 - (ii) Luminaire cleaning cycle specified.
 - (iii) Light source replacement strategy specified.
 - (iv) Maximum lumen depreciation allowable.

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- (v) Light source mortality allowable.
- (l) If required, the provision of power density indicator (PDI) and annual energy consumption indicator (AECI) calculated values for the lighting scheme.

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Appendix D (informative)

Historical (2005 edition) classification of luminaires their glare control and upward waste light

This Table is reproduced from Table 2.10 in the 2005 edition of this Standard (incorporating Amendment 1:2008). It has been reproduced here to provide the corresponding description of luminaires to those in the body of this Standard. It is provided for information only. The mandatory statements in the table are not requirements of this edition of the Standard but were normative in the 2005 edition.

1	2	3	4
Luminaire classification	Characteristics of luminaire light distribution	Glare control criteria ^a	Maximum UWLR ^b
Type 1	Approximately uniform light distribution in azimuth with maximum intensities at or near the horizontal A clear or translucent enclosure which neither disperses the lamp outline nor diminishes the brightness of the lamp (e.g. clear sphere)	See Note ^c	See Note ^c
Type 2	Similar light distribution to Type 1 but the outline of the lamp is either dispersed or shielded from direct view, providing an enclosure of a reasonably uniform brightness (e.g. opal or capped sphere)	The average luminance of the enclosure shall not exceed 10 kcd/m ² at (vertical) angles of between 70° and 90°. See Note ^e	40 % See Note ^d
Type 3	Approximately uniform light distribution in azimuth with maximum intensities directed below the horizontal. Like Type 2, the outline of the lamp is either dispersed or shielded from direct view, providing an enclosure of a reasonably uniform brightness (e.g. capped lantern style)	Where the outline of the lamp is dispersed above $\gamma = 75^\circ$, as specified for Type 2. Where any part of the lamp or reflector is visible above $\gamma = 75^\circ$ the intensity shall be not more than— (i) 270 cd/1000 lamp lumens at a (vertical) angle of 80°; and (ii) 120 cd/1000 lamp lumens at a (vertical) angle of 90°, at any angle of azimuth.	20 %
Type 4	An asymmetric light distribution in azimuth in which— (a) the maximum intensities are directed below the horizontal; and (b) in directions along the road to either side of the luminaire.	The intensity at any angle of azimuth shall be not more than— (i) 720 cd in total or 180 cd/1000 lamp lumens, whichever is the greater, at a vertical angle of 80°; and (ii) 300 cd in total or 80 cd/1000 lamp lumens, whichever is the greater, at a vertical angle of 90°.	8 %

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1	2	3	4
Luminaire classification	Characteristics of luminaire light distribution	Glare control criteria ^a	Maximum UWLR ^b
Type 5	Luminaire with a symmetrical or asymmetric beam distribution, with fixed mounting or capable of being adjusted with respect to the direction of aiming (e.g. Cat. V luminaire or aimable floodlight)	The absolute intensity shall be not more than— (a) 1000 cd at a (vertical) angle of 90°; and (b) 2500 cd at a (vertical) angle of 80°, at any angle of azimuth.	3 %
Type 6	Luminaire with symmetrical or asymmetric beam with flat horizontal visor and fixed mounting not capable of vertical aiming, e.g. road luminaire or floodlight both with flat horizontal visor ^f	Zero cd at γ (vertical) angle of 90° at any angle of azimuth.	0 %

^a The requirements apply to luminaire Types 1-4 in the position photometered and to Type 5 (aimable) as installed. All are based on the initial (100 h) lamp output.

^b The requirements apply to each luminaire in the attitude in which it is installed.

^c Type 1 luminaires produce excessive glare and upward waste light and are not suitable for use in the applications covered by this Standard.

^d Type 2 luminaires produce considerable glare and upward waste light and are not encouraged for use in the applications covered by this Standard.

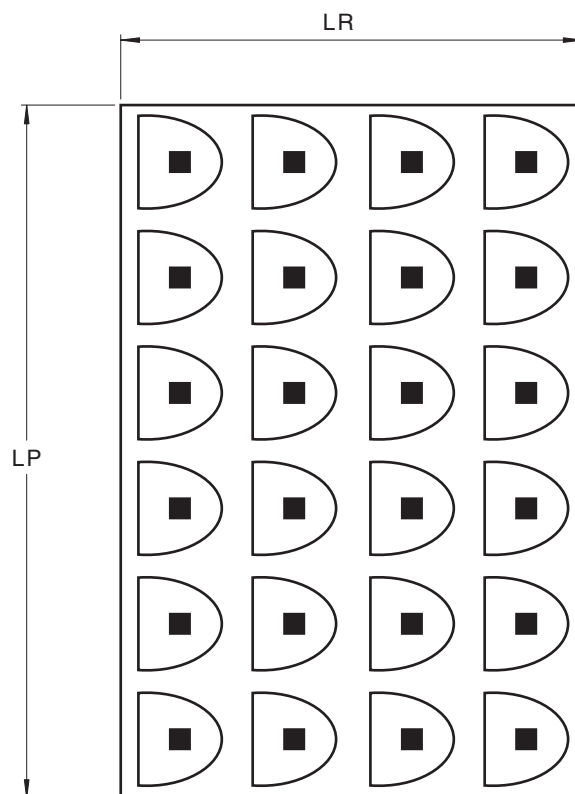
^e Compliance with glare control criteria may be determined by measurement with a luminance photometer or by finding the maximum value of the quotient of the absolute intensity at γ angles between 70° and 90° divided by the projected surface area of light emitting portions at that γ angle, i.e. finding the maximum value of $I(\gamma)/A(\gamma)$ between $\gamma = 70^\circ$ to 90° over all azimuth angles.

^f Refer to appropriate appendix in AS/NZS 4282 for illustration of typical examples.

Appendix E (normative)

Calculation of discomfort glare index

The luminous area of an LED luminaire is that area that encompasses the LED array and optics devices as shown in [Figures E1](#) and [E2](#).



Luminous area (LA) calculation
 $LA = (LP \times LR) \text{ m}^2$

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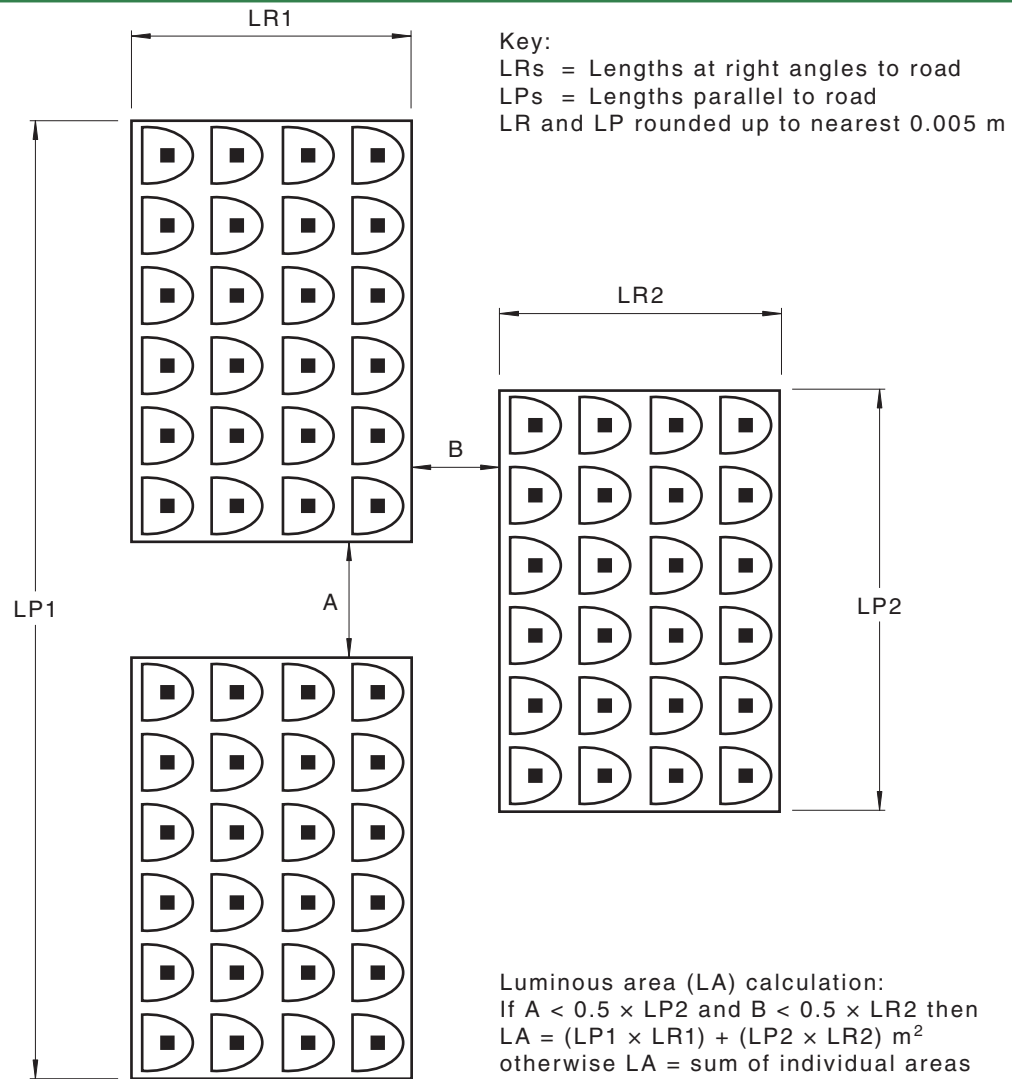
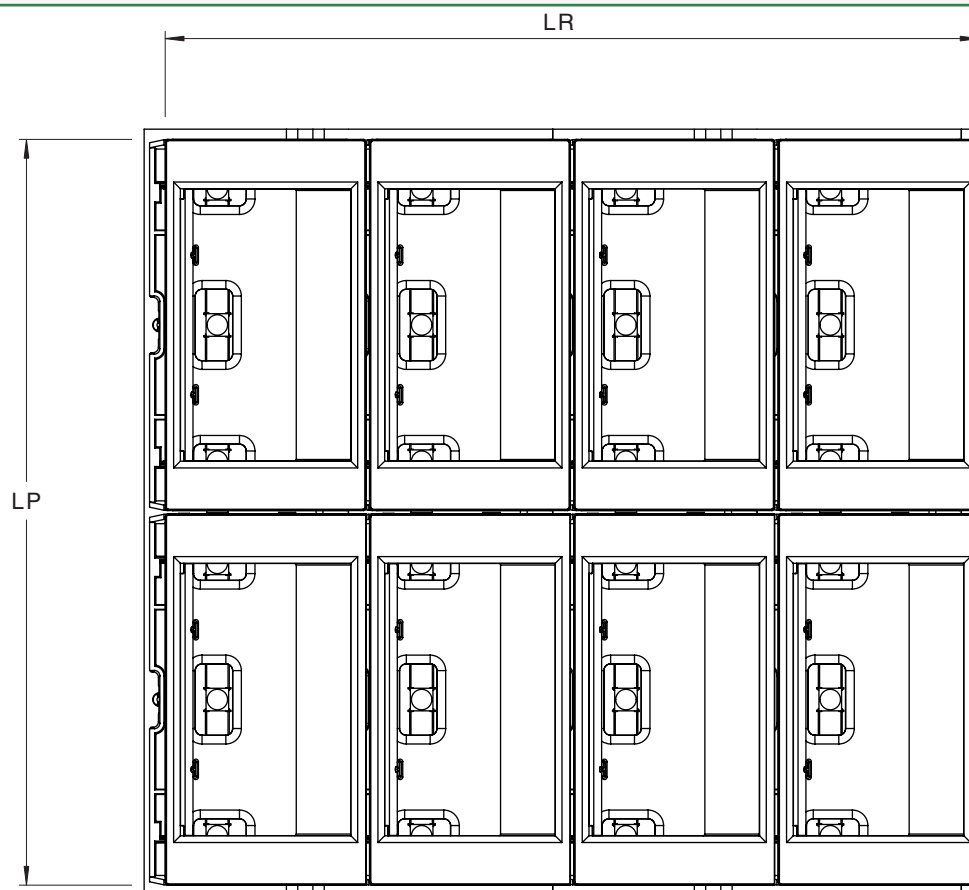


Figure E.1 — Dimension of luminous arrays to be used to calculate DGI values

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Luminous area (LA) calculation
 $LA = (LP \times LR) \text{ m}^2$

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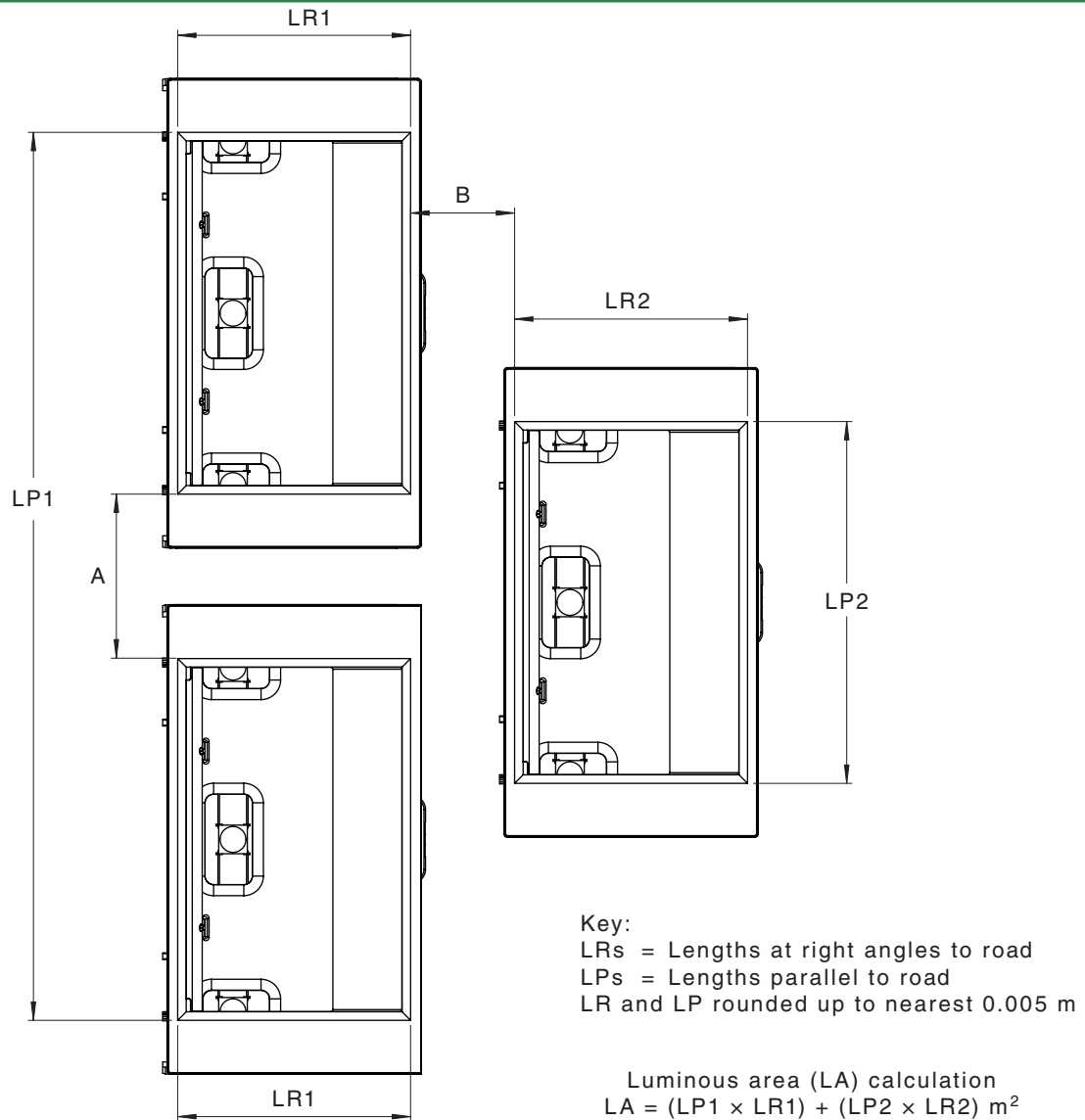
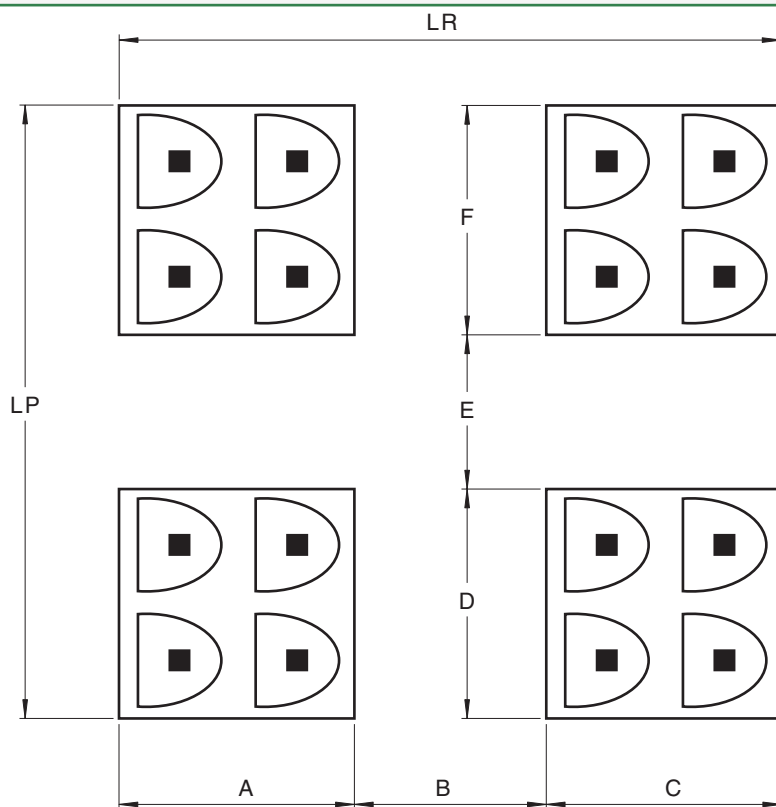


Figure E.2 — Dimension of luminous arrays to be used to calculate DGI values

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Luminous area (LA) calculation
 $LA = (LP \times LR) \text{ m}^2$

Key:

LRs = Lengths at right angles to road

LPs = Lengths parallel to road

LR and LP rounded up to nearest 0.005 m

Figure E.3 — Dimension of luminous arrays to be used to calculate DGI values — Any style of optics (reflector or lens or other)

Appendix F (normative)

Energy performance indicators

F.1 General

Where energy performance assessment and reporting is required, the parameters of power density indicator (PDI) and annual energy consumption indicator (AECI) shall be assessed according to the following methods and formulae.

NOTE PDI and AECI are compound parameters, thus they should be always presented together. All values and assumptions taken for the calculation of energy performance indicators should also be displayed clearly with the indicators. In some cases graphical depiction of the operational profile can be a suitable means of presentation.

F.2 Power density indicator (PDI)

F.2.1 General

Power density indicator for an area divided into sub-areas for a given state of operation shall be calculated with the following formula:

$$D_P = \frac{P}{\sum_{i=1}^n (\bar{E}_i \cdot A_i)} \quad \text{F1}$$

where

D_P = power density indicator, in watts per lux per square metre ($\text{W} \cdot \text{lx}^{-1} \cdot \text{m}^{-2}$)

P = system power of the lighting scheme used to light the relevant area, in watts (W)

$\bar{E}_i$ = maintained average horizontal illuminance of the sub-area 'i' determined in accordance with Paragraph F2.2, in lux (lx)

A_i = size of the sub-area 'i' lit by the lighting scheme, in square metres (m^2)

n = number of sub-areas to be lit

If the required lighting subcategory changes during the night and/or through the seasons (e.g. reductions in lighting subcategory due to decreased traffic density, changes in the visual environment or other relevant parameters), the power density (D_P) should be calculated separately for each of the lighting subcategories. Alternatively, where multiple lighting subcategories are used during the night or year the power density (D_P) may be calculated as an average over this period. The calculation shall clearly indicate the assumptions used for calculation of the power density (D_P) and how this value was evaluated.

Values of the power density indicator (D_P) shall be always presented and used together with the annual energy consumption indicator (D_E) for assessment of the energy performance of a particular lighting system.

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F.2.2 Average horizontal illuminance to be used for calculation of the power density indicator

For illuminance based lighting subcategories (i.e. Category P) the maintained average horizontal illuminance ($\bar{E}_h$) to be used for the power density (D_p) calculation shall be calculated according to [Clause 4.1.1](#).

From an energy efficiency and environmental perspective, the calculated lighting level for any lighting installation should not exceed the required lighting level of the next higher lighting subcategory (or not exceed the required lighting level by 50 % in case of the highest subcategory) without considering other design solutions.

F.2.3 System power for calculation of the power density indicator

The system power (P) shall be calculated from the sum of the operational power of the light sources, control gear and any other electrical devices which are directly associated with the lighting of the area to be lit. The system power (P) shall be calculated for either the complete lighting scheme or the representative section used during the lighting design according to the following formula:

$$P = \sum_{k=1}^{n_{lp}} P_k + P_{ad} \quad \text{F2}$$

where

P = total system power of the lighting scheme or its representative section, in watts (W)

P_k = operational power of the ' k 'th lighting point (light source, control gear, or any other device), in watts (W)

P_{ad} = total operational power of any devices not considered in P_k but necessary for the operation of the scheme such as a remote switch or photoelectric cell, or central management system, etc., in watts (W)

Where the system power is calculated for a representative area, the total operational power P_{ad} should be proportioned according to the number of luminaires used to illuminate the area over the total number of luminaires supplied from the devices represented by P_{ad} .

n_{lp} = number of lighting points associated with the lighting scheme or the representative section whichever is used in the calculation

If the lighting subcategory changes during the night and/or seasons, the system power (P) corresponding to the required lighting subcategory in that period should be calculated.

NOTE PDI can be a single number for full-time constant power operation and for 100 % dimming level in adaptive controlled systems, or it can represent different numbers for each considered state of operation.

Where the power to the luminaire is varied to maintain a constant luminous flux output throughout lifetime of the light sources [e.g. if constant light output (CLO) control gear is used], the average system power associated with these variations should be used for the calculation of system power (P) and density (D_p). The calculation shall clearly indicate the lifetime assumptions used for calculation of the average consumption and how this value was evaluated.

If the calculation for the main lighting subcategory is based on a single calculation for a section of the road, then the system power (P) calculation shall include the sum of the power of all luminaires and electrical devices which are inside and on the edges of the calculation area relevant to this scheme, in accordance with this Standard.

If the calculation is performed for an irregular shaped area, the system power (P) calculation shall include the sum of the power of each luminaire and electrical device which is needed to light the area.

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The system power (P) should not include any power associated with devices which are not used to fulfil the lighting function, even if they are connected to the same network.

F.2.4 Area (A_i) to be used for calculation of the power density indicator

The area used for calculation of the power density indicator (D_p) shall be identical to the area used in lighting design for lighting calculation of parameters according to [Clause 4.1.1](#) and described in AS/NZS 1158.2.

F.2.5 Power density indicator (PDI)—Calculation worked examples

F.2.5.1 Worked example of a lighting scheme without constant light output (CLO)

A local area road designed to AS/NZS 1158.3.1 Category P, with LED luminaires with off-peak adaptive lighting via central management system (CMS) controls.

Peak demand operation state to be in accordance with subcategory PR3 (required maintained average horizontal illuminance 1.75 lx), and off-peak demand operation state to be in accordance with subcategory PR5 (required maintained average horizontal illuminance 0.85 lx) (e.g. midnight to 6 a.m., say 0.5 of operating hours).

Lighting scheme physical parameters (numerical values are indicative only):

- (a) Road length = 1000 m.
- (b) Road reserve width (RRW) = 18 m.
- (c) Luminaire spacing = 50 m.
- (d) Luminaire quantity = 20.
- (e) Luminaire power (light source and light source control gear) = 20 W.
- (f) Light point controller power (e.g. NEMA/ANSI C136.41 device) = 2 W, one per luminaire.
- (g) System device power (e.g. RF controls communication gateway) = 30 W, one per scheme.

In accordance with Paragraph F2.1 above, where the lighting subcategory changes during the night, the power density (D_p) should be calculated separately for each of the lighting subcategories, and may be calculated as an average over this period.

The system power (P) calculation for the above example is—

$$P = \sum_{k=1}^{n_{lp}} P_k + P_{ad} \quad \text{F3}$$

where

P = total system power of the lighting scheme (W)

P_k = operational power of light point = $(20 + 2) = 22$ W

P_{ad} = total operational power of system devices = 30 W

Therefore the system power (P) = $(20 \times 22) + (30) = 470$ W.

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The power density indicator calculation (D_p) for the scheme for PR3 operation is—

$$D_p = \frac{P}{\sum_{i=1}^n (\bar{E}_i \cdot A_i)} \quad \text{F4}$$

where

D_p = power density indicator, in watts per lux per square metre ($\text{W.lx}^{-1}.\text{m}^{-2}$)

P = system power = 470 W

$\bar{E}_i$ = calculated maintained average horizontal illuminance is 1.79 lx (say) (for subcategory PR3)

A_i = size of the sub-area (1000×18) = 18 000 m^2

n = number of sub-areas to be lit = 1

Therefore the PR3 state power density indicator = $470 / (1.79 \times 18\,000) = \mathbf{0.01458 \text{ W.lx}^{-1}.\text{m}^{-2}}$.

The power density indicator calculation (D_p) for the scheme for PR5 operation is—

For PR5 state (required level is 0.85 lx) the dim level is approximately half compared with PR3 state (required level is 1.75 lx), thus for LED luminaires the luminaire power will also reduce by a factor of 0.5, to 10 W per unit.

The system power (P) calculation for the above example is—

$$P = \sum_{k=1}^{n_{lp}} P_k + P_{ad} \quad \text{F5}$$

where

P = total system power of the lighting scheme (W)

P_k = operational power of light point = $(10 + 2) = 12$ W

P_{ad} = total operational power of system devices = 30 W

Therefore the system power (P) = $(20 \times 12) + (30) = 270$ W.

The power density indicator calculation (D_p) for the scheme for PR5 operation is—

$$D_p = \frac{P}{\sum_{i=1}^n (\bar{E}_i \cdot A_i)} \quad \text{F6}$$

where

D_p = power density indicator, in watts per lux per square metre ($\text{W.lx}^{-1}.\text{m}^{-2}$)

P = system power = **270 W**

$\bar{E}_i$ = maintained average horizontal illuminance is 0.89 lx (say) (for subcategory PR5)

A_i = size of the sub-area (1000×18) = 18 000 m^2

n = number of sub-areas to be lit = 1

Therefore the PR5 state power density indicator = $270 / (0.89 \times 18\,000) = \mathbf{0.01685 \text{ W.lx}^{-1}.\text{m}^{-2}}$.

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The peak/off-peak (PR3/PR5) operating state hours (off peak, midnight to 6 a.m.) are pro-rata apportioned by nightly time duration to achieve an overall power density indicator result. In this particular case of equal time duration the two part-night power density indicators are averaged to achieve an overall power density indicator result.

$$D_p = (0.01458 + 0.01685) / 2 = 0.01571 \text{ W.lx}^{-1}.\text{m}^{-2}.$$

F.2.5.2 Worked example of a lighting scheme with constant light output (CLO)

This example uses the same physical and lighting design parameters as in the example above.

If the designed lighting scheme for PR3 operation uses a luminaire wattage of 20 W to achieve subcategory conformance at the end of the maintenance period; and if the lumen depreciation at the end of the maintenance period is, say 0.3 (L70), with CLO the initial luminaire wattage will be 14 W ($20 \text{ W} \times 0.7$).

The luminaire wattage appreciation curve over the maintenance period is the converse of the lumen depreciation curve. If the luminaire is assumed to have a straight lumen depreciation curve, the average luminaire wattage over the maintenance period is $(20 + 14)/2 = 17 \text{ W}$.

For PR5 operation with CLO the initial luminaire wattage will be 7 W ($10 \text{ W} \times 0.7$). The average luminaire wattage over the maintenance period is $(10 + 7)/2 = 8.5 \text{ W}$.

Alternatively, use the calculated lumen depreciation figures from IESNA LM-80-15 and IESNA TM-21-11 extrapolation including luminaire *in situ* life calculations for the luminaire concerned.

The system power (P) calculation for the above PR3 example with CLO is—

$$P = \sum_{k=1}^{n_{lp}} P_k + P_{ad} \quad \text{F7}$$

where

P = total system power of the lighting scheme (W)

P_k = operational power of light point = $(17 + 2) = 19 \text{ W}$

P_{ad} = total operational power of system devices = 30 W

Therefore the system power (P) = $(20 \times 19) + (30) = 410 \text{ W}$.

The power density indicator calculation (D_p) for the scheme for PR3 with CLO operation is—

$$D_p = \frac{P}{\sum_{i=1}^n (\bar{E}_i \cdot A_i)} \quad \text{F8}$$

where

D_p = power density indicator, in watts per lux per square metre ($\text{W.lx}^{-1}.\text{m}^{-2}$)

P = system power = 410 W

$\bar{E}_i$ = maintained average horizontal illuminance is 1.79 lx (say) (for subcategory PR3)

A_i = size of the sub-area (1000×18) = 18 000 m^2

n = number of sub-areas to be lit = 1

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Therefore the PR3 state with CLO power density indicator = $410/(1.79 \times 18\,000) = \mathbf{0.01273\,W.lx^{-1}.m^{-2}}$.

The system power (P) calculation for the above PR5 example with CLO operation is—

$$P = \sum_{k=1}^{n_p} P_k + P_{ad} \quad \text{F9}$$

where

P = total system power of the lighting scheme (W)

P_k = operational power of light point = $(8.5 + 2) = 10.5\,W$

P_{ad} = total operational power of system devices = $30\,W$

Therefore the system power (P) = $(20 \times 10.5) + (30) = 240\,W$.

The power density indicator calculation (D_p) for the scheme for PR5 with CLO operation is—

$$D_p = \frac{P}{\sum_{i=1}^n (\bar{E}_i \cdot A_i)} \quad \text{F10}$$

where

D_p = power density indicator, in watts per lux per square metre ($W.lx^{-1}.m^{-2}$)

P = system power = $240\,W$

$\bar{E}_i$ = maintained average horizontal illuminance is $0.89\,lx$ (say) (for subcategory PR5)

A_i = size of the sub-area $(1000 \times 18) = 18\,000\,m^2$

n = number of sub-areas to be lit = 1

Therefore the PR5 state with CLO power density indicator = $240/(0.89 \times 18\,000) = \mathbf{0.01498\,W.lx^{-1}.m^{-2}}$.

F.3 Annual energy consumption indicator (AECI)

F.3.1 General

The annual electricity consumption of a road lighting scheme depends on—

- the period of time for which lighting is provided;
- the lighting subcategory specified by the relevant lighting standard for each lighting period;
- the efficiency of the lighting scheme, when providing the necessary lighting for each period;
- the way the lighting management system follows the change in visual needs of road users; and
- the networked standby mode energy consumption of lighting devices during the period when the lighting is not needed (see Paragraph F2.3).

For comparison and monitoring of the energy performance of a lighting scheme, the energy consumption indicator shall take into account the annual accumulated energy use of road lighting illuminating the

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street or public area; however, actual lighting needs may vary during the year because of the following reasons:

- (i) Seasonal variations of daylight/night time hours; this depends on the geographical location of the area.
- (ii) Changing weather conditions which influence perceived visual performance (i.e. dry or wet road surface).
- (iii) Changing traffic density on the street or public area during the night (i.e. different temporal pattern of usage such as increased usage at 'rush hour') or following the fluctuation in social activity (i.e. school terms, national holiday periods).
- (iv) Changing functionality of the street or public area (i.e. roads are closed for a certain period or turned to pedestrian areas in festive seasons).

Annual energy consumption indicator (AECI) shall be calculated with the following formula:

$$D_E = \frac{\sum_{j=1}^m (P_j \cdot t_j)}{A} \quad \text{F11}$$

where

D_E = annual energy consumption indicator for a road lighting installation, in watt hours per square metre (Wh.m⁻²)

P_j = operational power associated with the j^{th} period of operation, in watts (W)

t_j = duration of j^{th} period of operation profile when the power P_j is consumed, over a year, in hours (h)

A = size of the area lit by the same lighting arrangement, in m²

m = number of periods with different operational power P_j . m shall also consider the period over which the non-active power is consumed. This period would generally be the time when the lighting is not operational, i.e. daylight hours and any night time period when the lighting is not lit

If the light output of a light source is intended to be constant, but the power consumption of this light source (or other electrical devices) varies in time [e.g. if constant light output (CLO) drivers are used], the average power consumption over the anticipated lifetime shall be included in the calculation. The calculation shall clearly indicate the lifetime assumptions used for calculation of the average consumption and how this value was evaluated.

Annual energy consumption indicator (D_E) shall be always presented and used together with the values of the power density indicator (D_P) for assessment of the energy performance of a particular lighting system.

F.3.2 Annual energy consumption indicator (AECI)—Calculation worked examples

F.3.2.1 Worked example without constant light output (CLO)

The following are worked examples of AECI calculations to show how the method is used in practice.

Example lighting scheme from Paragraph F2.5 above, with added lighting scheme operational parameters (numerical values are indicative only):

- (a) Light point controller power = 2 W, 24/7 applicability.

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- (b) System device power = 30 W, 24/7 applicability.
- (c) Night time operational power at PR3 = $(20 \times 22) + (30) = 470 \text{ W}$.
- (d) Day time operational power at PR3 = $(20 \times 2) + (30) = 70 \text{ W}$.
- (e) Night time operational power at PR5 = $(20 \times 12) + (30) = 270 \text{ W}$.
- (f) Day time operational power at PR5 = $(20 \times 2) + (30) = 70 \text{ W}$.
- (g) Night time duration of operation over a year = $365 \text{ days} \times 24 \text{ h} \times 0.5 = 4380 \text{ h/year}$.
- (h) Day time duration of operation over a year = $365 \text{ days} \times 24 \text{ h} \times 0.5 = 4380 \text{ h/year}$.

The annual energy consumption indicator (AECI) calculation is—

$$\text{A1} \quad D_E = [(470 \times 4380 \times 0.5) + (270 \times 4380 \times 0.5) + (70 \times 4380)] / 18\,000 = 107.07 \text{ Wh.m}^{-2} \quad \text{F12 A1}$$

where

D_E = annual energy consumption indicator (Wh.m^{-2})

P_j = operational power. Three states = PR3 470 W, PR5 270 W, day time 70 W

t_j = duration of operation over a year. Day time 4380 h, night time 4380 h

A = size of the area lit = $18\,000 \text{ m}^2$

m = number of periods with different operational power = 3

Therefore, $D_E = [(470 \times 4380) + (270 \times 4380) + (70 \times 4380)] / 18\,000 = 197.10 \text{ Wh.m}^{-2}$.

F3.2.2 Worked example with constant light output (CLO)

Example lighting scheme from Paragraph F2.5 above, with added lighting scheme operational parameters (numerical values are indicative only):

- (a) Light point controller power = 2 W, 24/7 applicability.
- (b) System device power = 30 W, 24/7 applicability.
- (c) Night time operational power at PR3 = $(20 \times 19) + (30) = 410 \text{ W}$.
- (d) Day time operational power at PR3 = $(20 \times 2) + (30) = 70 \text{ W}$.
- (e) Night time operational power at PR5 = $(20 \times 10.5) + (30) = 240 \text{ W}$.
- (f) Day time operational power at PR5 = $(20 \times 2) + (30) = 70 \text{ W}$.
- (g) Night time duration of operation over a year = $365 \text{ days} \times 24 \text{ h} \times 0.5 = 4380 \text{ h/year}$.
- (h) Day time duration of operation over a year = $365 \text{ days} \times 24 \text{ h} \times 0.5 = 4380 \text{ h/year}$.

The annual energy consumption indicator (AECI) calculation is—

$$\text{A1} \quad D_E = [(410 \times 4380 \times 0.5) + (240 \times 4380 \times 0.5) + (70 \times 4380)] / 18\,000 = 96.12 \text{ Wh.m}^{-2} \quad \text{F13 A1}$$

where

D_E = annual energy consumption indicator (Wh.m^{-2})

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P_j = operational power. Three states = PR3 410 W, PR5 240 W, day time 70 W

t_j = duration of operation over a year. Day time 4380 h, night time 4380 h

A = size of the area lit = 18 000 m²

m = number of periods with different operational power = 3

Therefore, $D_E = \left[(410 \times 4380) + (240 \times 4380) + (70 \times 4380) \right] / 18\,000 = 175.18 \text{ Wh} \cdot \text{m}^{-2}$.

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AS/NZS 2890.1, *Parking facilities, Part 1: Off-street car parking*

[A1] [Text deleted.] [A1]

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AS/NZS 60598.2.3, *Part 2.3: Particular requirements — Luminaires for road and street lighting*

AS/NZS ISO 31000, *Risk management—Guidelines*

SA/SNZ TS 1158.6, *Lighting for roads and public spaces, Part 6: Luminaires—Performance*

SA/SNZ HB 436, *Risk management guidelines—Companion to AS/NZS ISO 31000:2009*

EN 13201-2, *Road Lighting, Part 2: Performance requirements*

The following studies referred to in [Appendix A](#) provide further reading relating to lighting for fear of crime or enhancement of amenity:

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The reader’s attention is drawn to the following related documents:

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Amendment control sheet

AS 1158.3.1:2020 Amd 1:2024

Amendment No. 1 (2024)

Correction amendment

Summary: This amendment applies to the following:

- [Clauses 1.2, 1.3.5, 1.3.6, 2.4, 3.1.3.4, 3.2, 3.3, 3.4, 3.5.3, 3.6.1, 3.6.2, 3.7.1.2, 3.7.1.3, 4.2.2, 4.3.1, 4.3.2, 4.5.2.4, 4.6.1, 4.6.2, 4.6.3, 4.9.1, 4.9.2, F.3.2.1, F.3.2.2.](#)
- [Tables 2.2, 2.5, 3.1, 3.3, 3.4, 3.5, 3.6, 3.7, 3.8, 3.9, 3.10.](#)
- [Figures 2.1, 4.1, 4.2, 4.3, 4.4, 4.7, 4.8.](#)
- [Bibliography.](#)

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